

3. Stochastic Processes

We learn in kindergarten about the phenomenon of *Brownian motion*, the random jittery movement that a particle suffers when it is placed in a liquid. Famously, it is caused by the constant bombardment due to molecules in the surrounding the liquid. Our goal in this section is to introduce the mathematical formalism that allows us to model such random behaviour.

3.1 The Langevin Equation

In contrast to the previous section, we will here focus on just a single particle. However, this particle will be sitting in a background medium. If we know the force F acting on the particle, its motion is entirely deterministic, governed by

$$m\ddot{\vec{x}} = -\gamma\dot{\vec{x}} + \vec{F} \quad (3.1)$$

In contrast to the previous section, this is not a Hamiltonian system. This is because we have included a friction term with a coefficient γ . This arises due to the viscosity, η , of the surrounding liquid that we met in the previous section. If we model the particle as a sphere of radius a then there is a formula due to Stokes which says $\gamma = 6\pi\eta a$. However, in what follows we shall simply treat γ as a fixed parameter. In the presence of a time independent force, the steady-state solution with $\ddot{\vec{x}} = 0$ is

$$\dot{\vec{x}} = \frac{1}{\gamma} \vec{F}$$

For this reason, the quantity $1/\gamma$ is sometimes referred to as the *mobility*.

Returning to (3.1), for any specified force $\vec{F}$, the path of the particle is fully determined. This is seemingly at odds with the random behaviour observed in Brownian motion. The way in which we reconcile these two points is, hopefully, obvious: in Brownian motion the force $\vec{F}$ that the particle feels is itself random. In fact, we will split the force into two pieces,

$$\vec{F} = -\nabla V + \vec{f}(t)$$

Here V is a fixed background potential in which the particle is moving. Perhaps V arises because the particle is moving in gravity; perhaps because it is attached to a spring. But, either way, there is nothing random about V . In contrast, $\vec{f}(t)$ is the random force that the particle experiences due to all the other atoms in the liquid. It is sometimes referred to as *noise*. The resulting equation is called the *Langevin equation*

$$m\ddot{\vec{x}} = -\gamma\dot{\vec{x}} - \nabla V + \vec{f}(t) \quad (3.2)$$

Although it looks just like an ordinary differential equation, it is, in fact, a different beast known as a *stochastic differential equation*. The reason that it's different is that we don't actually know what $\vec{f}(t)$ is. Yet, somehow, we must solve this equation anyway!

Let's clarify what is meant by this. Suppose that you did know the microscopic force $\vec{f}(t)$ that is experienced by a given particle. Then you could, in principle, go ahead and solve the Langevin equation (3.2). But the next particle that you look at will experience a different force $\vec{f}(t)$ so you'll have to solve (3.2) again. And for the third particle, you'll have to solve it yet again. Clearly, this is going to become tedious. What's more, it's unrealistic to think that we will actually know $\vec{f}(t)$ in any specific case. Instead, we admit that we only know certain crude features of the force $\vec{f}(t)$ such as, for example, its average value. Then we might hope that this is sufficient information to figure out, say, the average value of $\vec{x}(t)$. That is the goal when solving the Langevin equation.

3.1.1 Diffusion in a Very Viscous Fluid

We start by solving the Langevin equation in the case of vanishing potential, $V = 0$. (For an arbitrary potential, the Langevin equation is an unpleasant non-linear stochastic differential equation and is beyond our ambition in this course. However, we will discuss some properties of the case with potential in the following section when we introduce the Fokker-Planck equation). We can simplify the problem even further by considering Brownian motion in a very viscous liquid. In this case, motion is entirely dominated by the friction term in the Langevin equation and we ignore the inertial term, which is tantamount to setting $m = 0$.

When $m = 0$, we're left with a first order equation,

$$\dot{\vec{x}}(t) = \frac{1}{\gamma} \vec{f}(t)$$

For any $\vec{f}(t)$, this can be trivially integrated to give

$$\vec{x}(t) = \vec{x}(0) + \frac{1}{\gamma} \int_0^t dt' \vec{f}(t') \quad (3.3)$$

At this point, we can't go any further until we specify some of the properties of the noise $\vec{f}(t)$. Our first assumption is that, on average, the noise vanishes at any given time. We will denote averages by $\langle \cdot \rangle$, so this assumption reads

$$\langle \vec{f}(t) \rangle = 0 \quad (3.4)$$

Taking the average of (3.3) then gives us the result:

$$\langle \vec{x}(t) \rangle = \vec{x}(0)$$

This is deeply unsurprising: if the average noise vanishes, the average position of the particle is simply where we left it to begin with. Nonetheless, it's worth stressing that this doesn't mean that all particles sit where you leave them. It means that if you drop many identical particles at the origin, $\vec{x}(0) = \vec{0}$, then they will all move but their average position — or their centre of mass — will remain at the origin.

We can get more information by looking at the variance of the position,

$$\langle (\vec{x}(t) - \vec{x}(0))^2 \rangle$$

This will tell us the average spread of the particles. We can derive an expression for the variance by first squaring (3.3) and then taking the average,

$$\langle (\vec{x}(t) - \vec{x}(0))^2 \rangle = \frac{1}{\gamma^2} \int_0^t dt'_1 \int_0^t dt'_2 \langle \vec{f}(t'_1) \cdot \vec{f}(t'_2) \rangle \quad (3.5)$$

In order to compute this, we need to specify more information about the noise, namely its correlation function $\langle f_i(t_1) f_j(t_2) \rangle$ where we have resorted to index notation, $i, j = 1, 2, 3$ to denote the direction of the force. This is specifying how likely it is that the particle will receive a given kick f_j at time t_2 given that it received a kick f_i at time t_1 .

In many cases of interest, including that of Brownian motion, the kicks imparted by the noise are both fast and uncorrelated. Let me explain what this means. Suppose that a given collision between our particle and an atom takes time τ_{coll} . Then if we focus on time scales less than τ_{coll} then there will clearly be a correlation between the forces imparted on our particle because these forces are due to the same process that's already taking place. (If an atom is coming in from the left, then it's still coming in from the left at a time $t \ll \tau_{\text{coll}}$ later). However if we look on time scales $t \gg \tau_{\text{coll}}$, the force will be due to a different collision with a different atom. The statement that the noise is uncorrelated means that the force imparted by later collisions knows nothing about earlier collisions. Mathematically, this means

$$\langle f_i(t_1) f_j(t_2) \rangle = 0 \quad \text{when} \quad t_2 - t_1 \gg \tau_{\text{coll}}$$

The statement that the collisions are fast means that we *only* care about time scales $t_2 - t_1 \gg \tau_{\text{coll}}$ and so can effectively take the limit $\tau_{\text{coll}} \rightarrow 0$. However, that doesn't

quite mean that we can just ignore this correlation function. Instead, when we take the limit $\tau_{\text{coll}} \rightarrow 0$, we're left with a delta-function contribution,

$$\langle f_i(t_1) f_j(t_2) \rangle = 2D\gamma^2 \delta_{ij} \delta(t_2 - t_1) \quad (3.6)$$

Here the factor of γ^2 has been put in for convenience. We will shortly see the interpretation of the coefficient D , which governs the strength of the correlations. Noise which obeys (3.4) and (3.6) is often referred to as *white noise*. It is valid whenever the environment relaxes back down to equilibrium much faster than the system of interest. This guarantees that, although the system is still reeling from the previous kick, the environment remembers nothing of what went before and kicks again, as fresh and random as the first time.

Using this expression for white noise, the variance (3.5) in the position of the particles is

$$\langle (\vec{x}(t) - \vec{x}(0))^2 \rangle = 6Dt \quad (3.7)$$

This is an important result: the root-mean square of the distance increases as $\sqrt{t}$ with time. This is characteristic behaviour of diffusion. The coefficient D is called the *diffusion constant*. (We put the factor of γ^2 in the correlation function (3.6) so that this equation would come out nicely).

3.1.2 Diffusion in a Less Viscous Liquid

Let's now return to the Langevin equation (3.2) and repeat our analysis, this time retaining the inertia term, so $m \neq 0$. We will still set $V = 0$.

As before, computing average quantities — this time both velocity $\langle \dot{\vec{x}}(t) \rangle$ and position $\langle \vec{x}(t) \rangle$ is straightforward and relatively uninteresting. For a given $\vec{f}(t)$, it is not difficult to solve (3.2). After multiplying by an integrating factor $e^{\gamma t/m}$, the equation becomes

$$\frac{d}{dt} (\dot{\vec{x}} e^{\gamma t/m}) = \frac{1}{m} \vec{f}(t) e^{\gamma t/m}$$

which can be happily integrated to give

$$\dot{\vec{x}}(t) = \dot{\vec{x}}(0) e^{-\gamma t/m} + \frac{1}{m} \int_0^t dt' \vec{f}(t') e^{\gamma(t'-t)/m} \quad (3.8)$$

We now use the fact that the average of noise vanishes (3.4) to find that the average velocity is simply that of a damped particle in the absence of any noise,

$$\langle \dot{\vec{x}}(t) \rangle = \dot{\vec{x}}(0) e^{-\gamma t/m}$$

Similarly, to determine the average position we have

$$\vec{x}(t) = \vec{x}(0) + \int_0^t dt' \dot{\vec{x}}(t') \quad (3.9)$$

From which we get

$$\begin{aligned} \langle \vec{x}(t) \rangle &= \vec{x}(0) + \int_0^t dt' \langle \dot{\vec{x}}(t') \rangle \\ &= \vec{x}(0) + \frac{m}{\gamma} \dot{\vec{x}}(0) (1 - e^{-\gamma t/m}) \end{aligned}$$

Again, this is unsurprising: when the average noise vanishes, the average position of the particle coincides with that of a particle that didn't experience any noise.

Things get more interesting when we look at the expectation values of quadratic quantities. This includes the variance in position $\langle \vec{x}(t) \cdot \vec{x}(t) \rangle$ and velocity $\langle \dot{\vec{x}}(t) \cdot \dot{\vec{x}}(t) \rangle$, but also more general correlation functions in which the two quantities are evaluated at different times. For example, the correlation function $\langle \dot{x}_i(t_1) \dot{x}_j(t_2) \rangle$ tells us information about the velocity of the particle at time t_2 given that we know where its velocity at time t_1 . From (3.8), we have the expression,

$$\langle \dot{x}_i(t_1) \dot{x}_j(t_2) \rangle = \langle \dot{x}_i(t_1) \rangle \langle \dot{x}_j(t_2) \rangle + \frac{1}{m^2} \int_0^{t_1} dt'_1 \int_0^{t_2} dt'_2 \langle f_i(t'_1) f_j(t'_2) \rangle e^{\gamma(t'_1+t'_2-t_1-t_2)/m}$$

where we made use of the fact that $\langle \vec{f}(t) \rangle = 0$ to drop the terms linear in the noise $\vec{f}$. If we use the white noise correlation function (3.6), and assume $t_2 \geq t_1 > 0$, the integral in the second term becomes,

$$\begin{aligned} \langle \dot{x}_i(t_1) \dot{x}_j(t_2) \rangle &= \langle \dot{x}_i(t_1) \rangle \langle \dot{x}_j(t_2) \rangle + \frac{2D\gamma^2}{m^2} \delta_{ij} e^{-\gamma(t_1+t_2)/m} \int_0^{t_1} dt' e^{2\gamma t'/m} \\ &= \langle \dot{x}_i(t_1) \rangle \langle \dot{x}_j(t_2) \rangle + \frac{D\gamma}{m} \delta_{ij} (e^{-\gamma(t_2-t_1)/m} - e^{-\gamma(t_1+t_2)/m}) \end{aligned}$$

For very large times, $t_1, t_2 \rightarrow \infty$, we can drop the last term as well as the average velocities since $\langle \dot{\vec{x}}(t) \rangle \rightarrow 0$. We learn that the correlation between velocities decays exponentially as

$$\langle \dot{x}_i(t_1) \dot{x}_j(t_2) \rangle \rightarrow \frac{D\gamma}{m} \delta_{ij} e^{-\gamma(t_2-t_1)/m}$$

This means that if you know the velocity of the particle at some time t_1 , then you can be fairly confident that it will have a similar velocity at a time $t_2 < t_1 + m/\gamma$ later. But if you wait longer than time m/γ then you would be a fool to make any bets on the velocity based only on your knowledge at time t_1 .

Finally, we can also use this result to compute the average velocity-squared (which, of course, is the kinetic energy of the system). At late times, the any initial velocity has died away and the resulting kinetic energy is due entirely to the bombardment by the environment. It is independent of time and given by

$$\langle \dot{\vec{x}}(t) \cdot \dot{\vec{x}}(t) \rangle = \frac{3D\gamma}{m} \quad (3.10)$$

One can compute similar correlation functions for position $\langle x_i(t_1)x_j(t_2) \rangle$. The expressions are a little more tricky but still quite manageable. (Combining equations (3.9) and (3.8), you can see that you will have a quadruple integral to perform and figuring out the limits is a little fiddly). At late times, it turns out that the variance of the position is given by the same expression that we saw for the viscous liquid (3.7),

$$\langle (\vec{x}(t) - \vec{x}(0))^2 \rangle = 6Dt \quad (3.11)$$

again exhibiting the now-familiar $\sqrt{t}$ behaviour for the root-mean-square distance.

3.1.3 The Einstein Relation

We brushed over something important and lovely in the previous discussion. We computed the average kinetic energy of a particle in (3.10). It is

$$E = \frac{1}{2}m\langle \dot{\vec{x}} \cdot \dot{\vec{x}} \rangle = \frac{3}{2}D\gamma$$

But we already know what the average energy of a particle is when it's bombarded by its environment: it is given by the equipartition theorem and, crucially, depends only on the temperature of the surroundings

$$E = \frac{3}{2}k_B T$$

It must be therefore that the diffusion constant D is related to the mobility $1/\gamma$ by

$$D = \frac{k_B T}{\gamma} \quad (3.12)$$

That's rather surprising! The diffusion constant captures the amount a particle is kicked around due to the background medium; the mobility expresses the how hard it is for a particle to plough through the background medium. And yet they are related. This equation is telling us that diffusion and viscosity both have their microscopic origin in the random bombardment of molecules. Notice that D is inversely proportional to γ . Yet you might have thought that the amount the particle is kicked increases as the viscosity increases. Indeed, looking back at (3.6), you can see that the amount the particle is kicked is actually proportional to $D\gamma^2 \sim T\gamma$. Which is more in line with our intuition.

Equation (3.12) is known as the Einstein relation. It is an important example of the fluctuation-dissipation theorem. The fluctuations of the particle as it undergoes its random walk are related to the drag force (or dissipation of momentum) that the particle feels as it moves through the fluid.

The Einstein relation gives an excellent way to determine Boltzmann's constant experimentally. Watch a particle perform a Brownian jitter. After time t , the distance travelled by the particle (3.7) should be

$$\langle \vec{x}^2 \rangle = \frac{k_B T}{\pi \eta a} t$$

where we have used the Stokes formula $\gamma = 6\pi\eta a$ to relate the mobility to the viscosity μ and radius a of the particle. This experiment was done in 1909 by the French physicist Jean Baptiste Perrin and won him the 1926 Nobel prize.

3.1.4 Noise Probability Distributions

So far, we've only needed to use the two pieces of information about the noise, namely,

$$\langle \vec{f}(t) \rangle = 0 \tag{3.13}$$

$$\langle f_i(t_1) f_j(t_2) \rangle = 2D\gamma^2 \delta_{ij} \delta(t_1 - t_2) \tag{3.14}$$

However, if we wanted to compute correlation functions involving more than two velocities or positions, it should be clear from the calculation that we would need to know higher moments of the probability distribution for $\vec{f}(t)$. In fact, the definition of *white noise* is that there are no non-trivial correlations other than $\langle f_i(t_1) f_j(t_2) \rangle$. This doesn't mean that the higher correlation functions are vanishing, just that they can be reduced to the two-time correlators. This means that for N even,

$$\langle f_{i_1}(t_1) \dots f_{i_N}(t_N) \rangle = \langle f_{i_1}(t_1) f_{i_2}(t_2) \rangle \dots \langle f_{i_{N-1}}(t_{N-1}) f_{i_N}(t_N) \rangle + \text{permutations}$$

while, for N odd

$$\langle f_{i_1}(t_1) \dots f_{i_N}(t_N) \rangle = 0$$

Another way of saying this is that all but the second cumulant of the probability distribution vanish.

Instead of specifying all these moments of the distribution, it is often much more useful to specify the probability distribution for $\vec{f}(t)$ directly. However, this is a slightly subtle object because we want to specify the probability for an entire function $\vec{f}(t)$, rather than a single random variable. This means that the probability distribution must be a functional: you give it a function $\vec{f}(t)$ and it spits back a number which, in this case, should be between zero and one.

The good news is that, among the class of probability distributions over functions, the white noise distribution is by far the easiest! If we were dealing with a single random variable, the distribution that has only two-point correlators but no higher is the Gaussian. And, suitably generalised, this also works for our functional probability distribution. The probability distribution that gives white noise is

$$\text{Prob}[f(t)] = \mathcal{N} \exp \left(- \int_{-\infty}^{+\infty} dt \frac{\vec{f}(t) \cdot \vec{f}(t)}{4D\gamma^2} \right)$$

where $\mathcal{N}$ is a normalization factor which is needed to ensure that the sum over all probabilities gives unity. This “sum” is really a sum over all functions $\vec{f}(t)$ or, in other words, a functional integral. The normalization condition which fixes $\mathcal{N}$ is then

$$\int Df(t) \text{Prob}[f(t)] = 1 \tag{3.15}$$

With this probability distribution, all averaging over the noise can now be computed as a functional integral. If you have any function $g(x)$, then its average is

$$\langle g(x) \rangle = \mathcal{N} \int Df(t) g(x_f) e^{-\int dt \vec{f}^2/4D\gamma^2}$$

where the notation x_f means the solution to the Langevin equation in the presence of a fixed source f .

Let’s now show that the Gaussian probability distribution indeed reproduces the white noise correlations as claimed. To do this, we first introduce an object $Z[\vec{J}(t)]$ known as a *generating function*. We can introduce a generating function for any probability distribution, so let’s keep things general for now and later specialise to the Gaussian distribution.

$$Z[\vec{J}(t)] = \int Df(t) \text{Prob}[f(t)] \exp \left(\int_{-\infty}^{+\infty} dt \vec{J}(t) \cdot \vec{f}(t) \right)$$

This generating function is a functional: it is a function of any function $\vec{J}(t)$ that we care to feed it. By construction, $Z[0] = 1$, courtesy of (3.15).

As the notation Z suggests, the generating function has much in common with the partition function that we work with in a first course of [Statistical Mechanics](#). This is most apparent in the context of statistical field theories where the generating function is reminiscent of the partition function. Both are functional, or path, integrals. These objects are also important in quantum field theory where the names partition function and generating function are often used synonymously.

The function $\vec{J}$ that we have introduced is, in this context, really little more than a trick that allows us to encode all the correlation functions in $Z[\vec{J}]$. To see how this works. Suppose that we differentiate Z with respect to $\vec{J}$ evaluated at some time $t = t_1$ and then set $\vec{J} = 0$. We have

$$\left. \frac{\delta Z}{\delta J_i(t_1)} \right|_{\vec{J}=0} = \int Df(t) f_i(t_1) \text{Prob}[f(t)] = \langle f_i(t_1) \rangle$$

Playing the same game, first taking n derivatives, gives

$$\begin{aligned} \left. \frac{\delta^n Z}{\delta J_{i_1}(t_1) \delta J_{i_2}(t_2) \dots J_{i_n}(t_n)} \right|_{\vec{J}=0} &= \int Df(t) f_{i_1}(t_1) f_{i_2}(t_2) \dots f_{i_n}(t_n) \text{prob}[f(t)] \\ &= \langle f_{i_1}(t_1) f_{i_2}(2) \dots f_{i_n}(t_n) \rangle \end{aligned}$$

So we see that if we can compute $Z[\vec{J}]$, then successive correlation functions are simply the coefficients of a Taylor expansion in $\vec{J}$. This is particularly useful for the Gaussian distribution where the generating function is,

$$Z[\vec{J}(t)] = \mathcal{N} \int Df(t) \exp \left(- \int_{-\infty}^{+\infty} dt \frac{\vec{f}(t) \cdot \vec{f}(t)}{4D\gamma^2} - \vec{J}(t) \cdot \vec{f}(t) \right)$$

But this is nothing more than a Gaussian integral. (Ok, it's an infinite number of Gaussian integrals because it's a functional integral. But we shouldn't let that phase us). We can easily compute it by completing the square

$$Z[\vec{J}(t)] = \mathcal{N} \int Df(t) \exp \left(- \frac{1}{4D\gamma^2} \int_{-\infty}^{+\infty} dt \left[\vec{f}(t) - 2D\gamma^2 \vec{J}(t) \right]^2 - 4D^2\gamma^4 \vec{J}(t) \cdot \vec{J}(t) \right)$$

After the shift of variable, $\vec{f} \rightarrow \vec{f} - 2D\gamma^2 \vec{J}$, the integral reduces to (3.15), leaving behind

$$Z[\vec{J}(t)] = \exp \left(D\gamma^2 \int_{-\infty}^{+\infty} dt \vec{J}(t) \cdot \vec{J}(t) \right)$$

Now it is an easy matter to compute correlation functions. Taking one derivative, we have

$$\frac{\delta Z}{\delta J_i(t_1)} = 2D\gamma^2 J_i(t_1) Z[\vec{J}]$$

But this vanishes when we set $J = 0$, in agreement with our requirement (3.13) that the average noise vanishes. Taking a second derivative gives,

$$\frac{\delta^2 Z}{\delta J_i(t_1) \delta J_j(t_2)} = 2D\gamma^2 \delta_{ij} \delta(t_1 - t_2) Z[\vec{J}] + 4D^2\gamma^4 J_i(t_1) J_j(t_2) Z[\vec{J}]$$

Now setting $\vec{J} = 0$, only the first term survives and reproduces the white noise correlation (3.14). One can continue the process to see that all higher correlation functions are entirely determined by $\langle f_i f_j \rangle$.

3.1.5 Stochastic Processes for Fields

Finally, it's worth mentioning that Langevin-type equations are not restricted to particle positions. It is also of interest to write down stochastic processes for fields. For example, we may want to consider a time dependent process for some order parameter $m(\vec{r}, t)$, influenced by noise

$$\frac{\partial m}{\partial t} = c\nabla^2 m - am - 2bm^2 + f$$

where $f(\vec{r}, t)$ is a random field with correlations $\langle f \rangle = 0$ and

$$\langle f(\vec{r}_1, t_1) f(\vec{r}_2, t_2) \rangle \sim \delta^d(\vec{r}_1 - \vec{r}_2) \delta(t_1 - t_2)$$

A famous example of a stochastic process is provided by the fluctuating boundary between, say, a gas and a liquid. Denoting the height of the boundary as $h(\vec{r}, t)$, the simplest description of the boundary fluctuations is given by the Edwards-Wilkinson equation,

$$\frac{\partial h}{\partial t} = \nabla^2 h + f$$

A somewhat more accurate model is given by the Kardar-Parisi-Zhang equation,

$$\frac{\partial h}{\partial t} = \nabla^2 h + \lambda(\nabla h)^2 + f$$

We won't have anything to say about the properties of these equations in this course. An introduction can be found in the second book by Kardar.

3.2 The Fokker-Planck Equation

Drop a particle at some position, say $\vec{x}_0$ at time t_0 . If the subsequent evolution is noisy, so that it is governed by a stochastic Langevin equation, then we've got no way to know for sure where the particle will be. The best that we can do is talk about probabilities. We will denote the probability that the particle sits at $\vec{x}$ at time t as $P(\vec{x}, t; \vec{x}_0, t_0)$.

In the previous section we expressed our uncertainty in the position of the particle in terms of correlation functions. Here we shift perspective a little. We would like to ask: what probability distribution $P(\vec{x}, t; \vec{x}_0, t_0)$ would give rise to the same correlation functions that arose from the Langevin equation?

We should stress that we care nothing about the particular path $\vec{x}(t)$ that the particle took. The probability distribution over paths would be a rather complicated functional (rather like those we saw in Section 3.1.4). Instead we will ask the much simpler question of the probability that the particle sits at $\vec{x}$ at time t , regardless of how it got there.

It is simple to write down a formal expression for the probability distribution. Let's denote the solution to the Langevin equation for a given noise function $\vec{f}$ as $\vec{x}_f$. Of course, if we know the noise, then there is no uncertainty in the probability distribution for $\vec{x}$. It is simply $P(\vec{x}, t) = \delta(\vec{x} - \vec{x}_f)$. Now averaging over all possible noise, we can write the probability distribution as

$$P(\vec{x}, t) = \langle \delta(\vec{x} - \vec{x}_f) \rangle \quad (3.16)$$

In this section, we shall show that the $P(\vec{x}, t)$ obeys a simple partial differential equation known as the Fokker-Planck equation.

3.2.1 The Diffusion Equation

The simplest stochastic process we studied was a particle subject to random forces in a very viscous fluid. The Langevin equation is

$$\dot{\vec{x}}(t) = \frac{1}{\gamma} \vec{f}(t)$$

In Section 3.1.1 we showed that the average position of the particle remains unchanged: if $\vec{x}(t=0) = \vec{0}$ then $\langle \vec{x}(t) \rangle = \vec{0}$ for all t . But the variance of the particle undergoes a random walk (3.7),

$$\langle \vec{x}(t)^2 \rangle = 6Dt \quad (3.17)$$

For this simple case, we won't derive the probability distribution: we'll just write it down. The probability distribution that reproduces this variance: it is just a Gaussian

$$P(\vec{x}, t) = \left(\frac{1}{4\pi Dt} \right)^{3/2} e^{-\vec{x}^2/4Dt} \quad (3.18)$$

where the factor out front is determined by the normalization requirement that

$$\int d^3x P(x, t) = 1$$

for all time t . Note that there is more information contained in this probability distribution than just the variance (3.17). Specifically, all higher cumulants vanish. (This means, for example, that $\langle \vec{x}^3 \rangle = 0$ and $\langle \vec{x}^4 \rangle = 3\langle \vec{x}^2 \rangle$ and so on). But it simple to check that this is indeed what arises from the Langevin equation with white noise described in Section 3.1.4.

The probability distribution (3.18) obeys the *diffusion equation*,

$$\frac{\partial P}{\partial t} = D \nabla^2 P$$

This is the simplest example of a Fokker-Planck equation. However, for more complicated versions of the Langevin equation, we will have to work harder to derive the analogous equation governing the probability distribution P .

3.2.2 Meet the Fokker-Planck Equation

Let's now consider the more general stochastic process. We'll still work in the viscous limit for now, setting $m = 0$ so that we have a first order Langevin equation,

$$\gamma \dot{\vec{x}} = -\nabla V + \vec{f} \quad (3.19)$$

A quadratic V corresponds to a harmonic oscillator potential and the Langevin equation is not difficult to solve. (In fact, mathematically it is the same problem that we solved in Section 3.1.2. You just have to replace $\dot{\vec{x}} = \vec{v} \rightarrow \vec{x}$). Any other V gives rise to a non-linear stochastic equation (confusingly sometimes called “quasi-linear” in this context) and no general solution is available. Nonetheless, we will still be able to massage this into the form of a Fokker-Planck equation.

We begin by extracting some information from the Langevin equation. Consider a particle sitting at some point x at time t . If we look again a short time δt later, the particle will have moved a small amount

$$\delta \vec{x} = \dot{\vec{x}} \delta t = -\frac{1}{\gamma} \nabla V \delta t + \frac{1}{\gamma} \int_t^{t+\delta t} dt' \vec{f}(t') \quad (3.20)$$

Here we've taken the average value of the noise function, f , over the small time interval. However, we've assumed that the displacement of the particle $\delta \vec{x}$ is small enough so that we can evaluate the force ∇V at the original position $\vec{x}$. (It turns out that this is ok in the present context but there are often pitfalls in making such assumptions in the theory of stochastic processes. We'll comment on one such pitfall at the end of this Section). We can now compute the average. Because $\langle \vec{f}(t) \rangle = 0$, we have

$$\langle \delta \vec{x} \rangle = -\frac{1}{\gamma} \nabla V \delta t \quad (3.21)$$

The computation $\langle \delta x_i \delta x_j \rangle$ is also straightforward,

$$\begin{aligned} \gamma^2 \langle \delta x_i \delta x_j \rangle &= \langle \partial_i V \partial_j V \rangle \delta t^2 - \delta t \int_t^{t+\delta t} dt' \langle \partial_i V f_j(t') + \partial_j V f_i(t') \rangle \\ &\quad + \int_t^{t+\delta t} dt' \int_t^{t+\delta t} dt'' \langle f_i(t') f_j(t'') \rangle \end{aligned}$$

Both the first two terms are order δt^2 . However, using (3.6), one of the integrals in the third term is killed by the delta function, leaving just one integral standing. This ensures that the third term is actually proportional to δt ,

$$\langle \delta x_i \delta x_j \rangle = 2\delta_{ij} D \delta t + \mathcal{O}(\delta t^2) \quad (3.22)$$

We will ignore the terms of order δt^2 . Moreover, It is simple to see that all higher correlation functions are higher order in δt . For example, $\langle \vec{x}^4 \rangle \sim \delta t^2$. These too will be ignored.

Our strategy now is to construct a probability distribution that reproduces (3.21) and (3.22). We start by considering the conditional probability $P(\vec{x}, t + \delta t; \vec{x}', t)$ that the particle sits at $\vec{x}$ at time $t + \delta t$ given that, a moment earlier, it was sitting at $\vec{x}'$. From the definition (3.16) we can write this as

$$P(\vec{x}, t + \delta t; \vec{x}', t) = \langle \delta(\vec{x} - \vec{x}' - \delta \vec{x}) \rangle$$

where δx is the random variable here; it is the distance moved in time δt . Next, we do something that may look fishy: we Taylor expand the delta-function. If you're nervous about expanding a distribution in this way, you could always regulate the delta function in your favourite manner to turn it into a well behaved function. However, more pertinently, we will see that the resulting expression sits inside an integral where any offending terms make perfect sense. For now, we just proceed naively

$$P(\vec{x}, t + \delta t; \vec{x}', t) = \left(1 + \langle \delta x_i \rangle \frac{\partial}{\partial x'_i} + \frac{1}{2} \langle \delta x_i \delta x_j \rangle \frac{\partial^2}{\partial x'_i \partial x'_j} + \dots \right) \delta(\vec{x} - \vec{x}') \quad (3.23)$$

We have truncated at second order because we want to compare this to (3.27) and, as we saw above, $\langle \delta \vec{x} \rangle$ and $\langle \delta \vec{x}^2 \rangle$ are the only terms that are order δt .

We now have all the information that we need. We just have to compare (3.27) and (3.23) and figure out how to deal with those delta functions. To do this, we need one more trick. Firstly, recall that our real interest is in the evolution of the probability $P(\vec{x}, t; \vec{x}_0, t_0)$, given some initial, arbitrary starting position $\vec{x}(t = t_0) = \vec{x}_0$. There is an obvious property that this probability must satisfy: if you look at some intermediate time $t_0 < t' < t$, then the particle has to be somewhere. Written as an equation, this “has to be somewhere” property is called the Chapman-Kolmogorov equation

$$P(\vec{x}, t; \vec{x}_0, t_0) = \int_{-\infty}^{+\infty} d^3 \vec{x}' P(\vec{x}, t; \vec{x}', t') P(\vec{x}', t'; \vec{x}_0, t_0) \quad (3.24)$$

Replacing t by $t + \delta t$, we can substitute our expression (3.23) into the Chapman-Kolmogorov equation, and then integrate by parts so that the derivatives on the delta

function turn and hit $P(\vec{x}', t'; \vec{x}_0, t_0)$. The delta-function, now unattended by derivatives, kills the integral, leaving

$$P(\vec{x}, t + \delta t; \vec{x}_0, t_0) = P(\vec{x}, t; \vec{x}_0, t_0) - \frac{\partial}{\partial x_i} \left(\langle \delta x_i \rangle P(\vec{x}, t; \vec{x}_0, t_0) \right) + \frac{1}{2} \langle \delta x_i \delta x_j \rangle \frac{\partial^2}{\partial x_i \partial x_j} P(\vec{x}, t; \vec{x}_0, t_0) + \dots \quad (3.25)$$

Using our expressions for $\langle \delta x \rangle$ and $\langle \delta x \delta x \rangle$ given in (3.21) and (3.22), this becomes

$$P(\vec{x}, t + \delta t; \vec{x}_0, t_0) = P(\vec{x}, t; \vec{x}_0, t_0) + \frac{1}{\gamma} \frac{\partial}{\partial x_i} \left(\frac{\partial V}{\partial x_i} P(\vec{x}, t; \vec{x}_0, t_0) \right) \delta t + D \frac{\partial^2}{\partial x^2} P(\vec{x}, t; \vec{x}_0, t_0) \delta t + \dots \quad (3.26)$$

But we can also get a much simpler expression for the left-hand side simply by Taylor expanding with respect to time,

$$P(\vec{x}, t + \delta t; \vec{x}_0, t_0) = P(\vec{x}, t; \vec{x}_0, t_0) + \frac{\partial}{\partial t} P(\vec{x}, t; \vec{x}_0, t_0) \delta t + \dots \quad (3.27)$$

Equating (3.27) with (3.26) gives us our final result,

$$\frac{\partial P}{\partial t} = \frac{1}{\gamma} \nabla \cdot (P \nabla V) + D \nabla^2 P \quad (3.28)$$

This is the *Fokker-Planck* equation. This form also goes by the name of the Smoluchowski equation or, for probabilists, Kolomogorov's forward equation.

Properties of the Fokker-Planck Equation

It is useful to write the Fokker-Planck equation as a continuity equation

$$\frac{\partial P}{\partial t} = -\nabla \cdot \vec{J} \quad (3.29)$$

where the *probability current* is

$$\vec{J} = -\frac{1}{\gamma} P \nabla V + D \nabla P \quad (3.30)$$

The second term is clearly due to diffusion (because there's a big capital D in front of it); the first term is due to the potential and is often referred to as the *drift*, meaning the overall motion of the particle due to background forces that we understand.

One advantage of writing the Fokker-Planck equation in terms of a current is that we see immediately that probability is conserved, meaning that if $\int d^3x P = 1$ at some point in time then it will remain so for all later times. This follows by a standard argument,

$$\frac{\partial}{\partial t} \int d^3x P = \int d^3x \frac{\partial P}{\partial t} = \int d^3x \nabla \cdot \vec{J} = 0$$

where the last equality follows because we have a total derivative (and we are implicitly assuming that there's no chance that the particle escapes to infinity so we can drop the boundary term).

The Fokker-Planck equation tells us how systems evolve. For some systems, such as those described by the diffusion equation, there is no end point to this evolution: the system just spreads out more and more. However, for generic potentials V there are time-independent solutions to the Fokker-Planck equation obeying $\nabla \cdot \vec{J} = 0$. These are the equilibrium configurations. The solution is given by

$$P(\vec{x}) \sim e^{-V(\vec{x})/\gamma D}$$

Using the Einstein relation (3.12), this becomes something very familiar. It is simply the Boltzmann distribution for a particle with energy $V(\vec{x})$ in thermal equilibrium

$$P(\vec{x}) \sim e^{-V(\vec{x})/k_B T} \quad (3.31)$$

Isn't that nice! (Note that there's no kinetic energy in the exponent as we set $m = 0$ as our starting point).

An Application: Escape over a Barrier

As an application of the Fokker-Planck equation, consider thermal escape from the one-dimensional potential shown in Figure 7. We'll assume that all the particles start off sitting close to the local minimum at $x_{\min}$. We model the potential close to this point as

$$V(x) \approx \frac{1}{2} \omega_{\min}^2 (x - x_{\min})^2$$

and we start our particles in a distribution that is effectively in local equilibrium (3.31), with

$$P(x, t = 0) = \sqrt{\frac{\omega_{\min}^2}{2\pi k_B T}} e^{-\omega_{\min}^2 (x - x_{\min})^2 / 2k_B T} \quad (3.32)$$

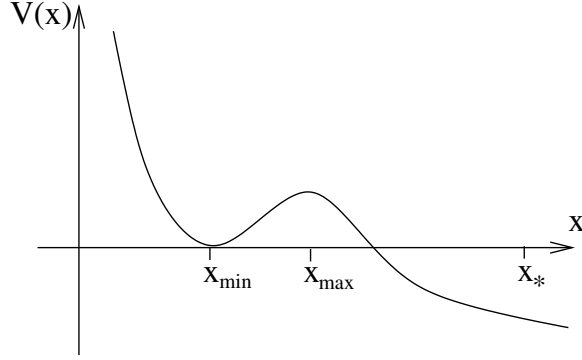


Figure 7: Escape over a Barrier.

But, globally, $x_{\min}$ is not the lowest energy configuration and this probability distribution is not the equilibrium configuration. In fact, as drawn, the potential has no global minimum and there is no equilibrium distribution. So this isn't what we'll set out to find. Instead, we would like to calculate the rate at which particles leak out of the trap and over the barrier.

Although we're clearly interested in a time dependent process, the way we proceed is to assume that the leakage is small and so can be effectively treated as a steady state process. This means that we think of the original probability distribution of particles (3.32) as a bath which, at least on the time scales of interest, is unchanging. The steady state leakage is modelled by a constant probability current $J = J_0$, with J given by (3.30). Using the Einstein relation $D = k_B T / \gamma$, we can rewrite this as

$$J = -\frac{k_B T}{\gamma} e^{-V(x)/k_B T} \frac{\partial}{\partial x} (e^{+V(x)/k_B T} P)$$

The first step is to integrate $J_0 e^{+V(x)/k_B T}$ between the minimum $x_{\min}$ and some distance far from all the action, $x \gg x_{\max}$, which we may as well call $x = x_*$,

$$\int_{x_{\min}}^{x_*} dx J_0 e^{V(x)/k_B T} = -\frac{k_B T}{\gamma} [e^{V(x_*)/k_B T} P(x_*) - e^{V(x_{\min})/k_B T} P(x_{\min})]$$

But we can take the probability $P(x_*)$ to be vanishingly small compared to $P(x_{\min})$ given in (3.32), leaving us with

$$\int_{x_{\min}}^{x_*} dx J_0 e^{V(x)/k_B T} \approx \frac{k_B T}{\gamma} \sqrt{\frac{\omega_{\min}^2}{2\pi k_B T}} \quad (3.33)$$

Meanwhile, the integral on the left-hand-side is dominated by the maximum of the potential. Let's suppose that close to the maximum, the potential looks like

$$V(x) \approx V_{\max} - \frac{1}{2}\omega_{\max}^2(x - x_{\max})^2$$

Then we'll write the integral as

$$J_0 \int_{x_{\min}}^{x_{\star}} dx e^{V(x)/k_B T} \approx J_0 e^{V_{\max}/k_B T} \sqrt{\frac{2\pi k_B T}{\omega_{\max}^2}} \quad (3.34)$$

Combining the two expressions (3.33) and (3.34), we get our final result for the rate of escape over the barrier

$$J_0 \approx \frac{\omega_{\min}\omega_{\max}}{2\pi\gamma} e^{-V_{\max}/k_B T}$$

3.2.3 Velocity Diffusion

So far we've ignored the inertia term, setting $m = 0$. Let's now put it back in. We can start by setting the potential to zero, so that the Langevin equation is

$$m\ddot{\vec{x}} = -\gamma\dot{\vec{x}} + \vec{f}(t)$$

But, we can trivially rewrite this as a first order equation involving $\vec{v} = \dot{\vec{x}}$,

$$m\dot{\vec{v}} = -\gamma\vec{v} + \vec{f}(t)$$

This means that if we're only interested in the distribution over velocities, $P(\vec{v}, t)$, then we have exactly the same problem that we've just looked at, simply replacing $\vec{x} \rightarrow \vec{v}$ and $\gamma \rightarrow m$. (Actually, you need to be a little more careful. The diffusion constant D that appears in (3.28) was really $D\gamma^2/\gamma^2$ where the numerator arose from the noise correlator and the denominator from the $\gamma\dot{\vec{x}}$ term in the Langevin equation. Only the latter changes, meaning that this combination gets replaced by $D\gamma^2/m^2$). The resulting Fokker-Planck equation is

$$\frac{\partial P}{\partial t} = \frac{1}{m} \frac{\partial}{\partial \vec{v}} \cdot \left(\gamma P \vec{v} + \frac{D\gamma^2}{m} \frac{\partial P}{\partial \vec{v}} \right) \quad (3.35)$$

The equilibrium distribution that follows from this obeys $\partial P/\partial t = 0$, meaning

$$\frac{\partial P}{\partial \vec{v}} = -\frac{m}{D\gamma} P \vec{v} \quad \Rightarrow \quad P = \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-m\vec{v}^2/2k_B T}$$

where we've again used the Einstein relation $D\gamma = k_B T$. This, of course, is the Maxwell-Boltzmann distribution.

In fact, we can do better than this. Suppose that we start all the particles off at $t = 0$ with some fixed velocity, $\vec{v} = \vec{v}_0$. This means that the probability distribution is a delta-function, $P(\vec{v}, t = 0) = \delta^3(\vec{v} - \vec{v}_0)$. We can write down a full time-dependent solution to the Fokker-Planck equation (3.35) with this initial condition.

$$P(\vec{v}, t) = \left(\frac{m}{2\pi k_B T (1 - e^{-2\gamma t/m})} \right)^{3/2} \exp \left(-\frac{m(\vec{v} - \vec{v}_0 e^{-\gamma t/m})^2}{2k_B T (1 - e^{-2\gamma t/m})} \right)$$

As $t \rightarrow \infty$, we return to the Maxwell-Boltzmann distribution. But now this tells us how we approach equilibrium.

The Kramers-Chandrasekhar Fokker-Planck Equation

As our final example of a Fokker-Planck equation, we can consider the Langevin equation with both acceleration term and potential term,

$$m\ddot{\vec{x}} = -\gamma\dot{\vec{x}} - \nabla V + \vec{f}(t)$$

Now we are looking for a probability distribution over phase space, $P(\vec{x}, \dot{\vec{x}}, t)$. The right way to proceed is to write this as two first-order equations. The first of these is simply the definition of velocity $\vec{v} = \dot{\vec{x}}$. The second is the Langevin equation

$$m\dot{\vec{v}} = -\gamma\vec{v} - \nabla V + \vec{f}(t)$$

These can now be combined into a single Langevin equation for six variables. Once armed with this, we need only follow the method that we saw above to arrive at a Fokker-Planck equation for $P(\vec{x}, \vec{v}, t)$,

$$\left(\frac{\partial}{\partial t} + v^i \frac{\partial}{\partial x^i} \right) P = \frac{1}{m} \frac{\partial}{\partial v^i} \left(\gamma v^i P + P \frac{\partial V}{\partial x^i} \right) + \frac{D\gamma^2}{m^2} \frac{\partial^2 P}{\partial v^i \partial v^i} \quad (3.36)$$

This form of the Fokker-Planck equations is sometimes called the Kramers equation and sometimes called the Chandrasekhar equation.

Note that this equation is now capturing the same physics that we saw in the Boltzmann equation: the probability distribution $P(\vec{x}, \vec{v}, t)$ is the same object that we called $f_1(\vec{r}, \vec{p}; t)$ in Section 2. Moreover, it is possible to derive this form of the Fokker-Planck equation, starting from the Boltzmann equation describing a heavy particle in a surrounding bath of light particles. The key approximation is that in small time intervals δt , the momentum of the heavy particle only changes by a small amount. Looking back, you can see that this was indeed an assumption in the derivation of the Fokker-Planck equation in Section 3.2.2, but not in the derivation of the Boltzmann equation.

Integrating over Velocity

The equation (3.36) governing the probability distribution over phase space $P(\vec{x}, \vec{v}, t)$ looks very different from the Fokker-Planck equation governing the probability distribution over configuration space (3.28). Yet the related Langevin equations are simply related by setting $m = 0$ or, equivalently, looking at systems with large γ . How can we derive (3.28) from (3.36)?

The computation involves a careful expansion of (3.36) in powers of $1/\gamma$. Let's see how this works. Firstly, we use the Einstein relation to write $D\gamma = k_B T$, and rearrange the terms in (3.36) to become

$$\frac{\partial}{\partial v^i} \left(\frac{k_B T}{m^2} \frac{\partial}{\partial v^i} + \frac{v^i}{m} \right) P = \frac{1}{\gamma} \left(\frac{\partial}{\partial t} + v^i \frac{\partial}{\partial x^i} - \frac{1}{m} \frac{\partial V}{\partial x^i} \frac{\partial}{\partial v^i} \right) P \quad (3.37)$$

We're going to solve this perturbatively in $1/\gamma$, writing

$$P = P^{(0)} + \frac{1}{\gamma} P^{(1)} + \frac{1}{\gamma^2} P^{(2)} + \dots$$

As our first pass at this, we drop anything that has a $1/\gamma$, which mean that $P^{(0)}$ must be annihilated by the left-hand-side of (3.37). This is a simple differential equation, with solution

$$P^{(0)}(v, x, t) = e^{-mv^2/2k_B T} \phi^{(0)}(x, t)$$

for any function $\phi^{(0)}(x, t)$. Of course, the velocity dependence here is simply the Maxwell-Boltzmann distribution. To figure out what restrictions we have on $\phi^{(0)}$, we need to go to the next order in perturbation theory. Keeping terms of $\mathcal{O}(1/\gamma)$, the differential equation (3.37) becomes

$$\frac{\partial}{\partial v^i} \left(\frac{k_B T}{m^2} \frac{\partial}{\partial v^i} + \frac{v^i}{m} \right) P^{(1)} = \left(\frac{\partial}{\partial t} + v^i \frac{\partial}{\partial x^i} + \frac{v^i}{k_B T} \frac{\partial V}{\partial x^i} \right) \phi^{(0)} e^{-mv^2/2k_B T} \quad (3.38)$$

But this equation cannot be solved for arbitrary $\phi^{(0)}$. This is simplest to see if we just integrate both sides over $\int d^3v$: the left-hand-side is a total derivative and so vanishes. On the right-hand-side, only one term remains standing and this must vanish. It is

$$\frac{\partial \phi^{(0)}}{\partial t} = 0$$

So $\phi^{(0)} = \phi^{(0)}(x)$. With this constraint, the solution to (3.38) is, again, straightforward to write down. It is

$$P^{(1)}(x, v, t) = \left(-mv^i \frac{\partial \phi^{(0)}}{\partial x^i} - \frac{m}{k_B T} v^i \frac{\partial V}{\partial x^i} \phi^{(0)} + \phi^{(1)}(x, t) \right) e^{-mv^2/2k_B T}$$

At this point, it doesn't look like we're making much progress. We still don't know what $\phi^{(0)}(x)$ is and we've now had to introduce yet another arbitrary function, $\phi^{(1)}(x, t)$ which carries all the time dependence. Let's plug this once more into (3.37), now working to order $\mathcal{O}(1/\gamma^2)$. After a little bit of algebra, the equation becomes

$$\begin{aligned} \frac{\partial}{\partial v^i} \left(\frac{k_B T}{m^2} \frac{\partial}{\partial v^i} + \frac{v^i}{m} \right) P^{(2)} = & \left[-m v^i v^j \frac{\partial}{\partial x^i} \left(\frac{\partial}{\partial x^j} + \frac{1}{k_B T} \frac{\partial V}{\partial x^j} \right) \phi^{(0)} \right. \\ & + \frac{\partial V}{\partial x^i} \left(\delta_{ij} - \frac{m}{k_B T} v^i v^j \right) \left(\frac{\partial}{\partial x^j} + \frac{1}{k_B T} \frac{\partial V}{\partial x^j} \right) \phi^{(0)} \\ & \left. + \left(\frac{\partial}{\partial t} + v^i \frac{\partial}{\partial x^i} + \frac{v^i}{k_B T} \frac{\partial V}{\partial x^i} \right) \phi^{(1)} \right] e^{-m v^2 / 2 k_B T} \end{aligned}$$

Once again, there's a consistency condition that must be realised if this equation is to have a solution. Integrating over $\int d^3 v$, the left-hand-side is a total derivative and therefore vanishes. Any term linear in v on the right-hand-side also vanishes. But so too do the terms on the second line: you can check that the Gaussian integral over the δ_{ij} term exactly cancels the $v^i v^j$ term. The resulting consistency condition is

$$\frac{\partial \phi^{(1)}}{\partial t} = k_B T \frac{\partial}{\partial x^i} \left(\frac{\partial}{\partial x^i} - \frac{1}{k_B T} \frac{\partial V}{\partial x^i} \right) \phi^{(0)} \quad (3.39)$$

where the overall factor of $k_B T$ on the right-hand-side comes only arises when you do the Gaussian integral over $\int d^3 v$.

Now we're almost there. (Although it probably doesn't feel like it!). Collecting what we've learned, to order $\mathcal{O}(1/\gamma)$, the probability distribution over phase space takes the form

$$P(x, v, t) = \left(\phi^{(0)} - \frac{m v^i}{\gamma} \frac{\partial \phi^{(0)}}{\partial x^i} - \frac{m v^i}{\gamma k_B T} \frac{\partial V}{\partial x^i} \phi^{(0)} + \frac{\phi^{(1)}}{\gamma} \right) e^{-m v^2 / 2 k_B T}$$

But to make contact with the earlier form of the Fokker-Planck equation (3.28), we want a distribution over configuration space. We get this by simply integrating over velocities. We'll also denote the resulting probability distribution as $P(x, t)$, with only the arguments to tell us that it's a different object:

$$P(x, t) = \int d^3 v P(x, v, t) = \sqrt{\frac{2\pi k_B T}{m}} \left(\phi^{(0)}(x) + \frac{1}{\gamma} \phi^{(1)}(x, t) \right)$$

But now we can use the consistency condition (3.39) to compute $\partial P / \partial t$. Working only to order $\mathcal{O}(1/\gamma)$, this reads

$$\frac{\partial P}{\partial t} = \frac{k_B T}{\gamma} \frac{\partial}{\partial x^i} \left(\frac{\partial}{\partial x^i} + \frac{1}{k_B T} \frac{\partial V}{\partial x^i} \right) P$$

Which is precisely the Fokker-Planck equation (3.28) that we saw previously.

3.2.4 Path Integrals: Schrödinger, Feynman, Fokker and Planck

There is a close similarity between the Fokker-Planck equation and the Schrödinger equation in quantum mechanics. To see this, let's return to the first order Langevin equation

$$\dot{\vec{x}} = \frac{1}{\gamma} \left(-\nabla V + \vec{f} \right) \quad (3.40)$$

and the corresponding Fokker-Planck equation (3.28). We can change variables to

$$P(x, t) = e^{-V(x)/2\gamma D} \tilde{P}(x, t) \quad (3.41)$$

Substituting into the Fokker-Planck equation, we see that the rescaled probability $\tilde{P}$ obeys

$$\frac{\partial \tilde{P}}{\partial t} = D \nabla^2 \tilde{P} + \left(\frac{1}{2\gamma} \nabla^2 V - \frac{1}{4\gamma^2 D} (\nabla V)^2 \right) \tilde{P} \quad (3.42)$$

There are no first order gradients $\nabla \tilde{P}$; only $\nabla^2 \tilde{P}$. This form of the Fokker-Planck equation looks very similar to the Schrödinger equation.

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + U(\vec{x}) \psi$$

All that's missing is a factor of i on the left-hand-side. Otherwise, with a few trivial substitutions, the two equations look more or less the same. Note, however, that the relationship between the potentials is not obvious: if we want to relate the two equations, we should identify

$$U = -\frac{1}{2\gamma} \nabla^2 V + \frac{1}{4D\gamma^2} (\nabla V)^2 \quad (3.43)$$

The relationship between the evolution of quantum and classical probabilities is also highlighted in the path integral formulation. Recall that the Schrödinger equation can be reformulated in terms of function integrals, with the quantum amplitude for a particle to travel from $\vec{x} = \vec{x}_i$ at time $t = t_i$ to $\vec{x}_f$ at time t_f given by⁸.

$$\langle \vec{x}_f, t_f | \vec{x}_i, t_i \rangle = \mathcal{N} \int \mathcal{D}x(t) \exp \left(\frac{i}{\hbar} \int dt \frac{\dot{\vec{x}}^2}{2m} - U(\vec{x}) \right)$$

⁸A derivation of the path integral from the Schrödinger equation can be found in the lectures on [Classical Dynamics](#).

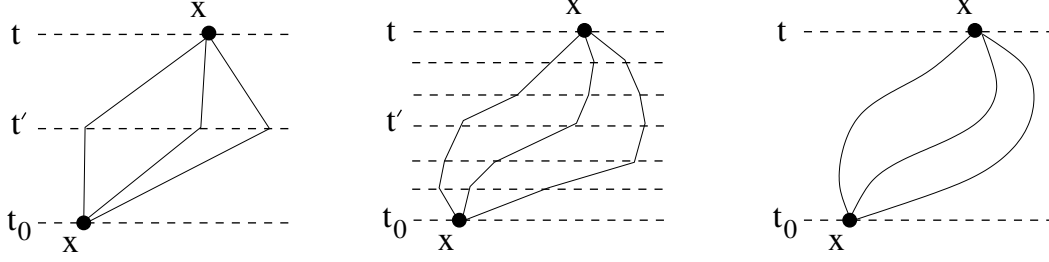


Figure 8: From Chapman-Kolmogorov to Feynman.

where $\mathcal{N}$ is a normalization factor. Here the integral is over all paths which start at $(\vec{x}_i, t_i)$ and end at $(\vec{x}_f, t_f)$. By analogy, we expect there to be a similar path integral formulation of the classical probability for a particle in the Langevin environment (3.40) to travel from $\vec{x}_i$ to $\vec{x}_f$. Indeed, the existence of a path integral formulation for this problem is very natural. The essence of this can already be seen in the Chapman-Kolmogorov equation (3.24)

$$P(\vec{x}, t; \vec{x}_0, t_0) = \int_{-\infty}^{+\infty} d^3 \vec{x}' P(\vec{x}, t; \vec{x}', t') P(\vec{x}', t'; \vec{x}_0, t_0)$$

This simply says that to get from point A to point B , a particle has to pass through some position in between. And we sum up the probabilities for each position. Adding many more intervening time steps, as shown in Figure 8, naturally suggests that we should be summing over all possible paths.

Deriving the Path Integral

Here we will sketch the derivation of the path integral formula for the Fokker-Planck equation. We've already met function integrals in Section 3.1.4 where we introduced the probability distribution for a given noise function $\vec{f}(t)$

$$\text{Prob}[f(t)] = \mathcal{N} \exp \left(- \int dt \frac{\vec{f}(t) \cdot \vec{f}(t)}{4D\gamma^2} \right) \quad (3.44)$$

subject to the normalization condition

$$\int \mathcal{D}f(t) \text{Prob}[f(t)] = 1 \quad (3.45)$$

But given a fixed noise profile $\vec{f}(t)$ and an initial condition, the path of the particle is fully determined by the Langevin equation (3.40). Let's call this solution $\vec{x}_f$. Then the

probability that the particle takes the path $\vec{x}_f$ is the same as the probability that the force is $\vec{f}$,

$$\begin{aligned}\text{Prob}[\vec{x}_f(t)] &= \text{Prob}[\vec{f}(t)] = \mathcal{N} \exp \left(- \int dt \frac{\vec{f}(t) \cdot \vec{f}(t)}{4D\gamma^2} \right) \\ &= \mathcal{N} \exp \left(- \frac{1}{4D\gamma^2} \int dt (\gamma \dot{\vec{x}}_f + \nabla V(\vec{x}_f))^2 \right)\end{aligned}$$

where, in the last line, we've used the Langevin equation (3.40) to relate the force to the path taken. But since this equation holds for any path $\vec{x}_f$, we can simply drop the f label. We have the probability that the particle takes a specific path $\vec{x}(t)$ given by

$$\text{Prob}[\vec{x}(t)] = \mathcal{N} \exp \left(- \frac{1}{4D\gamma^2} \int dt (\gamma \dot{\vec{x}} + \nabla V)^2 \right)$$

The total probability to go from $\vec{x}_i$ to $\vec{x}_f$ should therefore just be the sum over all these paths. With one, slightly fiddly, subtlety: the probability is normalized in (3.45) with respect to the integration measure over noise variable $\vec{f}$. And we want to integrate over paths. This means that we have to change integration variables and pick up a Jacobian factor for our troubles. We have

$$\begin{aligned}\text{Prob}[\vec{x}_f, t_f; \vec{x}_i, t_i] &= \mathcal{N} \int \mathcal{D}f(t) \exp \left(- \frac{1}{4D\gamma^2} \int dt (\gamma \dot{\vec{x}}_f + \nabla V(\vec{x}_f))^2 \right) \\ &= \mathcal{N} \int \mathcal{D}x(t) \det \mathcal{M} \exp \left(- \frac{1}{4D\gamma^2} \int dt (\gamma \dot{\vec{x}} + \nabla V)^2 \right) \quad (3.46)\end{aligned}$$

Here the operator $\mathcal{M}(t, t')$ that appears in the Jacobian be thought of as $\delta f(t)/\delta x(t')$. It can be written down by returning to the Langevin equation (3.40) which relates f and x ,

$$\mathcal{M}(t, t') = \gamma \frac{\partial}{\partial t} \delta(t - t') + \nabla^2 V \delta(t - t')$$

If we want to think in a simple minded way, we can consider this as a (very large) matrix $\mathcal{M}_{tt'}$, with columns labelled by the index t and rows labelled by t' . We'll write the two terms in this matrix as $\mathcal{M} = A + B$ so the determinant becomes

$$\det(A + B) = \det A \det(1 + A^{-1}B) \quad (3.47)$$

The first operator $A = \gamma \partial/\partial t \delta(t - t')$ doesn't depend on the path and its determinant just gives a constant factor which can be absorbed into the normalization $\mathcal{N}$. The operator A^{-1} in the second term is defined in the usual way as

$$\int dt' A(t, t') A^{-1}(t', t'') = \delta(t - t'')$$

where the integral over $\int dt'$ is simply summing over the rows of A and the columns of A^{-1} as in usual matrix multiplication. It is simple to check that the inverse is simply the step function

$$A^{-1}(t', t'') = \frac{1}{\gamma} \theta(t' - t'') \quad (3.48)$$

Now we write the second factor in (3.47) and expand,

$$\det(1 + A^{-1}B) = \exp\left(\text{Tr} \log(1 + A^{-1}B)\right) = \exp\left(\sum_n \text{Tr}(A^{-1}B)^n / n\right) \quad (3.49)$$

Here we should look in more detail at what this compact notation means. The term $\text{Tr} A^{-1}B$ is really short-hand for

$$\text{Tr} A^{-1}B = \int dt dt' A^{-1}(t, t') B(t', t)$$

where the integral over $\int dt'$ is multiplying the matrices together while the integral over $\int dt$ comes from taking the trace. Using (3.48) we have

$$\text{Tr} A^{-1}B = \frac{1}{\gamma} \int dt dt' \theta(t - t') \nabla^2 V \delta(t - t') = \frac{\theta(0)}{\gamma} \int dt \nabla^2 V$$

The appearance of $\theta(0)$ may look a little odd. This function is defined to be $\theta(x) = +1$ for $x > 0$ and $\theta(x) = 0$ for $x < 0$. The only really sensible value at the origin is $\theta(0) = 1/2$. Indeed, this follows from the standard regularizations of the step function, for example

$$\theta(x) = \lim_{\mu \rightarrow 0} \left(\frac{1}{2} + \frac{1}{\pi} \tan^{-1} \left(\frac{x}{\mu} \right) \right) \Rightarrow \theta(0) = \frac{1}{2}$$

What happens to the higher powers of $(A^{-1}B)^n$? Writing them out, we have

$$\begin{aligned} \text{Tr}(A^{-1}B)^n = \int dt \int dt_1 \dots dt_{2n-1} \theta(t - t_1) \delta(t_1 - t_2) \theta(t_2 - t_3) \delta(t_3 - t_4) \dots \\ \dots \theta(t_{2n-2} - t_{2n-1}) \delta(t_{2n-1} - t) \frac{(\nabla^2 V)^n}{\gamma^n} \end{aligned}$$

where we have been a little sloppy in writing $(\nabla^2 V)^n$ because each of these is actually computed at a different time. We can use the delta-functions to do half of the integrals, say all the t_n for n odd. We get

$$\text{Tr}(A^{-1}B)^n = \int dt \int dt_2 dt_4 \dots \theta(t - t_2) \theta(t_2 - t_4) \theta(t_4 - t_6) \dots \theta(t_{2n-2} - t) \frac{(\nabla^2 V)^n}{\gamma^n}$$

But this integral is only non-vanishing only if $t > t_2 > t_4 > t_6 > \dots > t_n > t$. In other words, the integral vanishes. (Note that you might think we could again get contributions from $\theta(0) = 1/2$, but the integrals now mean that the integrand has support on a set of zero measure. And with no more delta-functions to rescue us, gives zero. The upshot of this is that the determinant (3.49) can be expressed as a single exponential

$$\det(1 + A^{-1}B) = \exp\left(\frac{1}{2\gamma} \int dt \nabla^2 V\right)$$

We now have an expression for the measure factor in (3.46). Using this, the path integral for the probability becomes,

$$\begin{aligned} \text{Prob}[\vec{x}_f, t_f; \vec{x}_i, t_i] &= \mathcal{N}' \int \mathcal{D}x(t) \exp\left(-\frac{1}{4D\gamma^2} \int dt (\gamma\dot{\vec{x}} + \nabla V)^2 + \frac{1}{2\gamma} \int dt \nabla^2 V\right) \\ &= \mathcal{N}' e^{[V(x_f) - V(x_i)]/2\gamma D} \int \mathcal{D}x(t) \exp\left(-\int dt \frac{\dot{\vec{x}}^2}{4D} + U\right) \end{aligned}$$

where U is given in (3.43). Notice that the prefactor $e^{[V(x_f) - V(x_i)]/2\gamma D}$ takes the same form as the map from probabilities P to the rescaled $\tilde{P}$ in (3.41). This completes our derivation of the path integral formulation of probabilities.

3.2.5 Stochastic Calculus

There is one final generalization of the Langevin equation that we will mention but won't pursue in detail. Let's return to the case $m = 0$, but generalise the noise term in the Langevin equation so that it is now spatially dependent. We write

$$\gamma\dot{\vec{x}} = -\nabla V + b(\vec{x}) \vec{f}(t) \quad (3.50)$$

This is usually called the non-linear Langevin equation. The addition of the $b(\vec{x})$ multiplying the noise looks like a fairly innocuous change. But it's not. In fact, annoyingly, this equation is not even well defined!

The problem is that the system gets a random kick at time t , the strength of which depends on its position at time t . But if the system is getting a delta-function impulse at time t then its position is not well defined. Mathematically, this problem arises when we look at the position after some small time δt . Our equation (3.20) now becomes

$$\delta\vec{x} = \dot{\vec{x}} \delta t = -\frac{1}{\gamma} \nabla V \delta t + \frac{1}{\gamma} \int_t^{t+\delta t} dt' b(\vec{x}(t')) \vec{f}(t')$$

and our trouble is in making sense of the last term. There are a couple of obvious ways we could move forward:

- Ito: We could insist that the strength of the kick is related to the position of the particle immediately before the kick took place. Mathematically, we replace the integral with

$$\int_t^{t+\delta t} dt' b(\vec{x}(t')) \vec{f}(t') \longrightarrow b(\vec{x}(t)) \int_t^{t+\delta t} dt' \vec{f}(t')$$

This choice is known as *Ito stochastic calculus*.

- Stratonovich: Alternatively, we might argue that the kick isn't really a delta function. It is really a process that takes place over a small, but finite, time. To model this, the strength of the kick should be determined by the average position over which this process takes place. Mathematically, we replace the integral with,

$$\int_t^{t+\delta t} dt' b(\vec{x}(t')) \vec{f}(t') \longrightarrow \frac{1}{2} [b(\vec{x}(t + \delta t)) + b(\vec{x}(t))] \int_t^{t+\delta t} dt' \vec{f}(t')$$

This choice is known as *Stratonovich stochastic calculus*.

Usually in physics, issues of this kind don't matter too much. Typically, any way of regulating microscopic infinitesimals leads to the same macroscopic answers. However, this is not the case here and the Ito and Stratonovich methods give different answers in the continuum. In most applications of physics, including Brownian motion, the Stratonovich calculus is the right way to proceed because, as we argued when we first introduced noise, the delta-function arising in the correlation function $\langle f(t)f(t') \rangle$ is just a convenient approximation to something more smooth. However, in other applications such as financial modelling, Ito calculus is correct.

The subject of stochastic calculus is a long one and won't be described in this course. For the Stratonovich choice, the Fokker-Planck equation turns out to be

$$\frac{\partial P}{\partial t} = \frac{1}{\gamma} \nabla \cdot [P(\nabla V - D\gamma^2 b \nabla b)] + D \nabla^2 (b^2 P)$$

This is also the form of the Fokker-Planck equation that you get by naively dividing (3.50) by $b(\vec{x})$ and the defining a new variable $\dot{\vec{y}} = \dot{\vec{x}}/b$ which reduces the problem to our previous Langevin equation (3.19). In contrast, if we use Ito stochastic calculus, the $b \nabla b$ term is absent in the resulting Fokker-Planck equation.

4. Linear Response

The goal of response theory is to figure out how a system reacts to outside influences. These outside influences are things like applied electric and magnetic fields, or applied pressure, or an applied driving force due to some guy sticking a spoon into a quantum liquid and stirring.

We've already looked at a number of situations like this earlier in these lectures. If you apply a shearing force to a fluid, its response is to move; how much it moves is determined by the viscosity. If you apply a temperature gradient, the response is for heat to flow; the amount of heat is determined by the thermal conductivity. However, in both of these cases, the outside influence was time independent. Our purpose here is to explore the more general case of time dependent influences. As we'll see, by studying the response of the system at different frequencies, we learn important information about what's going on inside the system itself.

4.1 Response Functions

Until now, our discussion has been almost entirely classical. Here we want to deal with both classical and quantum worlds. For both cases, we start by explaining mathematically what is meant by an outside influence on a system.

Forces in Classical Dynamics

Consider a simple dynamical system with some generalized coordinates $x_i(t)$ which depend on time. If left alone, these coordinates will obey some equations of motion,

$$\ddot{x}_i + g_i(\dot{x}, x) = 0$$

This dynamics need not necessarily be Hamiltonian. Indeed, often we'll be interested in situations with friction. The outside influence in this example arises from perturbing the system by the addition of some driving forces $F_i(t)$, so that the equations of motion become,

$$\ddot{x}_i + g_i(\dot{x}, x) = F_i(t) \tag{4.1}$$

In this expression, $x_i(t)$ are dynamical degrees of freedom. This is what we're solving for. In contrast, $F_i(t)$ are *not* dynamical: they're forces that are under our control, like someone pulling on the end of a spring. We get to decide on the time dependence of each $F_i(t)$.

It may be useful to have an even more concrete example at the back of our minds. For this, we take every physicist's favorite toy: the simple harmonic oscillator. Here we'll include a friction term, proportional to γ , so that we have the damped harmonic oscillator with equation of motion

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = F(t) \quad (4.2)$$

We will discuss this model in some detail in section 4.2.

Sources in Quantum Mechanics

In quantum mechanics, we introduce the outside influences in a slightly different manner. The observables of the system are now operators, $\mathcal{O}_i$. We'll work in the Heisenberg picture, so that the operators are time dependent: $\mathcal{O} = \mathcal{O}(t)$. Left alone, the dynamics of these operators will be governed by a Hamiltonian $H(\mathcal{O})$. However, we have no interest in leaving the system alone. We want to give it a kick. Mathematically this is achieved by adding an extra term to the Hamiltonian,

$$H_{\text{source}}(t) = \phi_i(t)\mathcal{O}_i(t) \quad (4.3)$$

The $\phi_i(x)$ are referred to as *sources*. They are external fields that are under our control, analogous to the driving forces in the example above. Indeed, if we take a classical Hamiltonian and add a term of the form $x\phi$ then the resulting Euler-Lagrange equations include the source ϕ on the right-hand-side in the same way that the force F appears in (4.2).

4.1.1 Linear Response

We want to understand how our system reacts to the presence of the source or the driving force. To be concrete, we'll chose to work in the language of quantum mechanics, but everything that we discuss in this section will also carry over to classical systems. Our goal is to understand how the correlation functions of the theory change when we turn on a source (or sources) $\phi_i(x)$.

In general, it's a difficult question to understand how the theory is deformed by the sources. To figure this out, we really just need to sit down and solve the theory all over again. However, we can make progress under the assumption that the source is a small perturbation of the original system. This is fairly restrictive but it's the simplest place where we can make progress so, from now on, we focus on this limit. Mathematically, this means that we assume that the change in the expectation value of any operator is *linear* in the perturbing source. We write

$$\delta\langle\mathcal{O}_i(t)\rangle = \int dt' \chi_{ij}(t;t') \phi_j(t') \quad (4.4)$$

Here $\chi_{ij}(t; t')$ is known as a *response function*. We could write a similar expression for the classical dynamical system (4.1), where $\delta\langle\mathcal{O}_i\rangle$ is replaced by $x_i(t)$ and ϕ is replaced by the driving force $F_j(t)$. In classical mechanics, it is clear from the form of the equation of motion (4.1) that the response function is simply the Green's function for the system. For this reason, the response functions are often called Green's functions and you'll often see them denoted as G instead of χ .

From now on, we'll assume that our system is invariant under time translations. In this case, we have

$$\chi_{ij}(t; t') = \chi_{ij}(t - t')$$

and it is useful to perform a Fourier transform to work in frequency space. We define the Fourier transform of the function $f(t)$ to be

$$f(\omega) = \int dt e^{i\omega t} f(t) \quad \text{and} \quad f(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} f(\omega) \quad (4.5)$$

In particular, we will use the convention where the two functions are distinguished only by their argument.

Taking the Fourier transform of (4.4) gives

$$\begin{aligned} \delta\langle\mathcal{O}_i(\omega)\rangle &= \int dt' \int dt e^{i\omega t} \chi_{ij}(t - t') \phi_j(t') \\ &= \int dt' \int dt e^{i\omega(t-t')} \chi_{ij}(t - t') e^{i\omega t'} \phi_j(t') \\ &= \chi_{ij}(\omega) \phi_j(\omega) \end{aligned} \quad (4.6)$$

We learn the response is “local” in frequency space: if you shake something at frequency ω , it responds at frequency ω . Anything beyond this lies within the domain of non-linear response.

In this section we'll describe some of the properties of the response function $\chi(\omega)$ and how to interpret them. Many of these properties follow from very simple physical input. To avoid clutter, we'll mostly drop both the i, j indices. When there's something interesting to say, we'll put them back in.

4.1.2 Analyticity and Causality

If we work with a real source ϕ and a Hermitian operator $\mathcal{O}$ (which means a real expectation value $\langle\mathcal{O}\rangle$) then $\chi(t)$ must also be real. Let's see what this means for the

Fourier transform $\chi(\omega)$. It's useful to introduce some new notation for the real and imaginary parts,

$$\begin{aligned}\chi(\omega) &= \text{Re } \chi(\omega) + i\text{Im } \chi(\omega) \\ &\equiv \chi'(\omega) + i\chi''(\omega)\end{aligned}$$

This notation in terms of primes is fairly odd the first time you see it, but it's standard in the literature. You just have to remember that, in this context, primes do not mean derivatives!

The real and imaginary parts of the response function $\chi(\omega)$ have different interpretations. Let's look at these in turn

- **Imaginary Part:** We can write the imaginary piece as

$$\begin{aligned}\chi''(\omega) &= -\frac{i}{2}[\chi(\omega) - \chi^*(\omega)] \\ &= -\frac{i}{2} \int_{-\infty}^{+\infty} dt \chi(t)[e^{i\omega t} - e^{-i\omega t}] \\ &= -\frac{i}{2} \int_{-\infty}^{+\infty} dt e^{i\omega t}[\chi(t) - \chi(-t)]\end{aligned}$$

We see that the imaginary part of $\chi(\omega)$ is due to the part of the response function that is *not* invariant under time reversal $t \rightarrow -t$. In other words, $\chi''(\omega)$ knows about the arrow of time. Since microscopic systems are typically invariant under time reversal, the imaginary part $\chi''(\omega)$ must be arising due to dissipative processes.

$\chi''(\omega)$ is called the *dissipative* or *absorptive* part of the response function. It is also known as the *spectral function*. It will turn out to contain information about the density of states in the system that take part in absorptive processes. We'll see this more clearly in an example shortly.

Finally, notice that $\chi''(\omega)$ is an odd function,

$$\chi''(-\omega) = -\chi''(\omega)$$

- **Real Part:** The same analysis as above shows that

$$\chi'(\omega) = \frac{1}{2} \int_{-\infty}^{+\infty} dt e^{i\omega t}[\chi(t) + \chi(-t)]$$

The real part doesn't care about the arrow of time. It is called the *reactive* part of the response function. It is an even function,

$$\chi'(-\omega) = +\chi'(\omega)$$

Before we move on, we need to briefly mention what happens when we put the labels i, j back on the response functions. In this case, a similar analysis to that above shows that the dissipative response function comes from the anti-Hermitian part,

$$\chi''_{ij}(\omega) = -\frac{i}{2}[\chi_{ij}(\omega) - \chi_{ji}^*(\omega)] \quad (4.7)$$

Causality

We can't affect the past. This statement of causality means that any response function must satisfy

$$\chi(t) = 0 \quad \text{for all } t < 0$$

For this reason, χ is often referred to as the causal Green's function or *retarded Green's function* and is sometimes denoted as $G_R(t)$. Let's see what this simple causality requirement means for the Fourier expansion of χ ,

$$\chi(t) = \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \chi(\omega)$$

When $t < 0$, we can perform the integral by completing the contour in the upper-half plane (so that the exponent becomes $-i\omega \times (-i|t|) \rightarrow -\infty$). The answer has to be zero.

Of course, the integral is given by the sum of the residues inside the contour. So if we want the response function to vanish for all $t < 0$, it must be that $\chi(\omega)$ has no poles in the upper-half plane. In other words, causality requires:

$$\chi(\omega) \text{ is analytic for } \text{Im } \omega > 0$$

4.1.3 Kramers-Kronig Relation

The fact that χ is analytic in the upper-half plane means that there is a relationship between the real and imaginary parts, χ' and χ'' . This is called the Kramers-Kronig relation. Our task in this section is to derive it. We start by providing a few general mathematical statements about complex integrals.

A Discontinuous Function

First, consider a general function $\rho(\omega)$. We'll ask that $\rho(\omega)$ is meromorphic, meaning that it is analytic apart from at isolated poles. But, for now, we won't place any restrictions on the position of these poles. (We will shortly replace $\rho(\omega)$ by $\chi(\omega)$ which, as we've just seen, has no poles in the upper half plane). We can define a new function $f(\omega)$ by the integral,

$$f(\omega) = \frac{1}{i\pi} \int_a^b \frac{\rho(\omega')}{\omega' - \omega} d\omega' \quad (4.8)$$

Here the integral is taken along the interval $\omega' \in [a, b]$ of the real line. However, when ω also lies in this interval, we have a problem because the integral diverges at $\omega' = \omega$. To avoid this, we can simply deform the contour of the integral into the complex plane, either running just above the singularity along $\omega' + i\epsilon$ or just below the singularity along $\omega' - i\epsilon$. Alternatively (in fact, equivalently) we could just shift the position of the singularity to $\omega \rightarrow \omega \mp \epsilon$. In both cases we just skim by the singularity and the integral is well defined. The only problem is that we get different answers depending on which way we do things. Indeed, the difference between the two answers is given by Cauchy's residue theorem,

$$\frac{1}{2}[f(\omega + i\epsilon) - f(\omega - i\epsilon)] = \rho(\omega) \quad (4.9)$$

The difference between $f(\omega + i\epsilon)$ and $f(\omega - i\epsilon)$ means that the function $f(\omega)$ is discontinuous across the real axis for $\omega \in [a, b]$. If $\rho(\omega)$ is everywhere analytic, this discontinuity is a branch cut.

We can also define the average of the two functions either side of the discontinuity. This is usually called the *principal value*, and is denoted by adding the symbol $\mathcal{P}$ before the integral,

$$\frac{1}{2}[f(\omega + i\epsilon) + f(\omega - i\epsilon)] \equiv \frac{1}{i\pi} \mathcal{P} \int_a^b \frac{\rho(\omega')}{\omega' - \omega} d\omega' \quad (4.10)$$

We can get a better handle on the meaning of this principal part if we look at the real and imaginary pieces of the denominator in the integrand $1/[\omega' - (\omega \pm i\epsilon)]$,

$$\frac{1}{\omega' - (\omega \pm i\epsilon)} = \frac{\omega' - \omega}{(\omega' - \omega)^2 + \epsilon^2} \pm \frac{i\epsilon}{(\omega' - \omega)^2 + \epsilon^2} \quad (4.11)$$

By taking the sum of $f(\omega + i\epsilon)$ and $f(\omega - i\epsilon)$ in (4.10), we isolate the real part, the first term in (4.11). This is shown in the left-hand figure. It can be thought of as a suitably cut-off version of $1/(\omega' - \omega)$. It's as if we have deleted a small segment of this function lying symmetrically about divergent point ω and replaced it with a smooth function going through zero. This is the usual definition of the principal part of an integral.

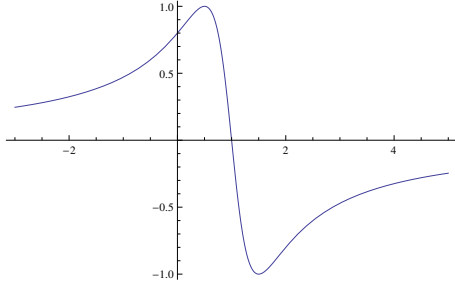


Figure 9: The real part of the function (4.11), plotted with $\omega' = 1$ and $\epsilon = 0.5$.

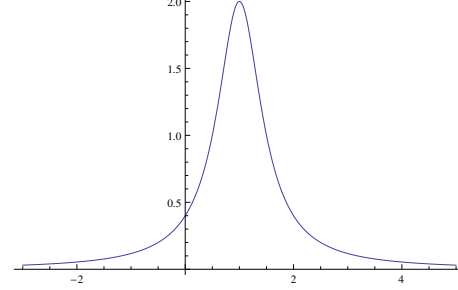


Figure 10: The imaginary part of the function (4.11), plotted with $\omega' = 1$ and $\epsilon = 0.5$

We can also see the meaning of the imaginary part of $1/(\omega' - \omega)$, the second term in (4.11). This is shown in the right-hand figure. As $\epsilon \rightarrow 0$, it tends towards a delta function, as expected from (4.9). For finite ϵ , it is a regularized version of the delta function.

Kramers-Kronig

Let's now apply this discussion to our response function $\chi(\omega)$. We'll be interested in the integral

$$\frac{1}{i\pi} \oint_C d\omega' \frac{\chi(\omega')}{\omega' - \omega} \quad \omega \in \mathbf{R} \quad (4.12)$$

where the contour C skims just above the real axis, before closing at infinity in the upper-half plane. We'll need to make one additional assumption: that $\chi(z)$ falls off faster than $1/|z|$ at infinity. If this holds, the integral is the same as we consider in (4.8) with $[a, b] \rightarrow [-\infty, +\infty]$. Indeed, in the language of the previous discussion, the integral is $f(\omega - i\epsilon)$, with $\rho = \chi$.

We apply the formulae (4.9) and (4.10). It gives

$$f(\omega - i\epsilon) = \frac{1}{i\pi} \mathcal{P} \left[\int_{-\infty}^{+\infty} d\omega' \frac{\chi(\omega')}{\omega' - \omega} \right] - \chi(\omega)$$

But we know the integral in (4.12) has to be zero since $\chi(\omega)$ has no poles in the upper-half plane. This means that $f(\omega - i\epsilon) = 0$, or

$$\chi(\omega) = \frac{1}{i\pi} \mathcal{P} \int_{-\infty}^{+\infty} d\omega' \frac{\chi(\omega')}{\omega' - \omega} \quad (4.13)$$

The important part for us is that factor of “ i ” sitting in the denominator. Taking real and imaginary parts, we learn that

$$\text{Re } \chi(\omega) = \mathcal{P} \int_{-\infty}^{+\infty} \frac{d\omega'}{\pi} \frac{\text{Im} \chi(\omega')}{\omega' - \omega} \quad (4.14)$$

and

$$\text{Im} \chi(\omega) = -\mathcal{P} \int_{-\infty}^{+\infty} \frac{d\omega'}{\pi} \frac{\text{Re} \chi(\omega')}{\omega' - \omega} \quad (4.15)$$

These are the *Kramers-Kronig* relations. They follow from causality alone and tell us that the dissipative, imaginary part of the response function $\chi''(\omega)$ is determined in terms of the reactive, real part, $\chi'(\omega)$ and vice-versa. However, the relationship is not local in frequency space: you need to know $\chi'(\omega)$ for all frequencies in order to reconstruct χ'' for any single frequency.

There’s another way of writing these relations which is also useful and tells us how we can reconstruct the full response function $\chi(\omega)$ if we only know the dissipative part. To see this, look at

$$\int_{-\infty}^{+\infty} \frac{d\omega'}{i\pi} \frac{\text{Im} \chi(\omega')}{\omega' - \omega - i\epsilon} \quad (4.16)$$

where the $-i\epsilon$ in the denominator tells us that this is an integral just below the real axis. Again using the formulae (4.9) and (4.10), we have

$$\begin{aligned} \int_{-\infty}^{+\infty} \frac{d\omega'}{i\pi} \frac{\text{Im} \chi(\omega')}{\omega' - \omega - i\epsilon} &= \text{Im} \chi(\omega) + \mathcal{P} \int_{-\infty}^{+\infty} \frac{d\omega'}{i\pi} \frac{\text{Im} \chi(\omega')}{\omega' - \omega - i\epsilon} \\ &= \text{Im} \chi(\omega) - i \text{Re} \chi(\omega) \end{aligned} \quad (4.17)$$

Or, rewriting as $\chi(\omega) = \text{Re} \chi(\omega) + i \text{Im} \chi(\omega)$, we get

$$\chi(\omega) = \int_{-\infty}^{+\infty} \frac{d\omega'}{\pi} \frac{\text{Im} \chi(\omega')}{\omega' - \omega - i\epsilon} \quad (4.18)$$

If you know the dissipative part of the response function, you know everything.

An Application: Susceptibility

Suppose that turning on a perturbation ϕ induces a response $\langle \mathcal{O} \rangle$ for some observable of our system. Then the *susceptibility* is defined as

$$\chi = \left. \frac{\partial \langle \mathcal{O} \rangle}{\partial \phi} \right|_{\omega=0}$$

We've called the susceptibility χ which is the same name that we gave to the response function. And, indeed, from the definition of linear response (4.4), the former is simply the zero frequency limit of the latter:

$$\chi = \lim_{\omega \rightarrow 0} \chi(\omega)$$

A common example, which we met in our first course in [Statistical Mechanics](#), is the change of magnetization M of a system in response to an external magnetic field B . The aptly named magnetic susceptibility is given by $\chi = \partial M / \partial B$.

From (4.18), we can write the susceptibility as

$$\chi = \int_{-\infty}^{+\infty} \frac{d\omega'}{\pi} \frac{\text{Im}\chi(\omega')}{\omega' - i\epsilon} \quad (4.19)$$

We see that if you can do an experiment to determine how much the system absorbs at all frequencies, then from this information you can determine the response of the system at zero frequency. This is known as the thermodynamic sum rule.

4.2 Classical Examples

The definitions and manipulations of the previous section can appear somewhat abstract the first time you encounter them. Some simple examples should shed some light. The main example we'll focus on is the same one that accompanies us through most of physics: the classical harmonic oscillator.

4.2.1 The Damped Harmonic Oscillator

The equation of motion governing the damped harmonic oscillator in the presence of a driving force is

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = F(t) \quad (4.20)$$

Here γ is the friction. We denote the undamped frequency as ω_0 , saving ω for the frequency of the driving force as in the previous section. We want to determine the response function, or Green's function, $\chi(t - t')$ of this system. This is the function which effectively solves the dynamics for us, meaning that if someone tells us the driving force $F(t)$, the motion is given by

$$x(t) = \int_{-\infty}^{+\infty} dt' \chi(t - t') F(t') \quad (4.21)$$

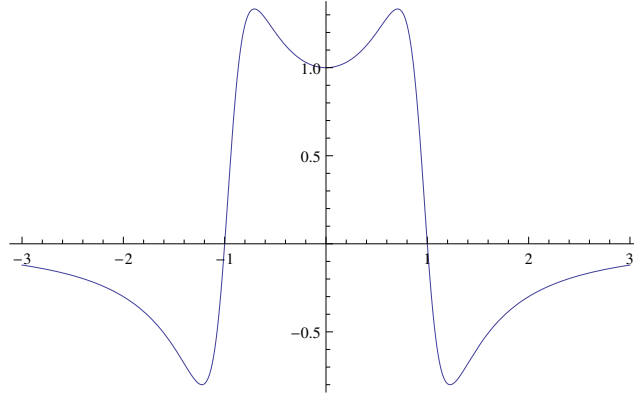


Figure 11: The real, reactive part of the response function for the underdamped harmonic oscillator, plotted with $\omega_0 = 1$ and $\gamma = 0.5$.

There is a standard method to figure out $\chi(t)$. Firstly, we introduce the (inverse) Fourier transform

$$\chi(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} \chi(\omega)$$

We plug this into the equation of motion (4.20) to get

$$\int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} \int_{-\infty}^{+\infty} dt' [-\omega^2 - i\gamma\omega + \omega_0^2] e^{-i\omega(t-t')} \chi(\omega) F(t') = F(t)$$

which is solved if the $\int d\omega$ gives a delta function. But since we can write a delta function as $2\pi\delta(t) = \int d\omega e^{-i\omega t}$, that can be achieved by simply taking

$$\chi(\omega) = \frac{1}{-\omega^2 - i\gamma\omega + \omega_0^2} \quad (4.22)$$

There's a whole lot of simple physics sitting in this equation which we'll now take some time to extract. All the lessons that we'll learn carry over to more complicated systems.

Firstly, we can look at the *susceptibility*, meaning $\chi(\omega = 0) = 1/\omega_0^2$. This tells us how much the observable changes by a perturbation of the system, i.e. a static force: $x = F/\omega_0^2$ as expected.

Let's look at the structure of the response function on the complex ω -plane. The poles sit at $\omega_*^2 + i\gamma\omega_* - \omega_0^2 = 0$ or, solving the quadratic, at

$$\omega_* = -\frac{i\gamma}{2} \pm \sqrt{\omega_0^2 - \gamma^2/4}$$

There are two different regimes that we should consider separately,

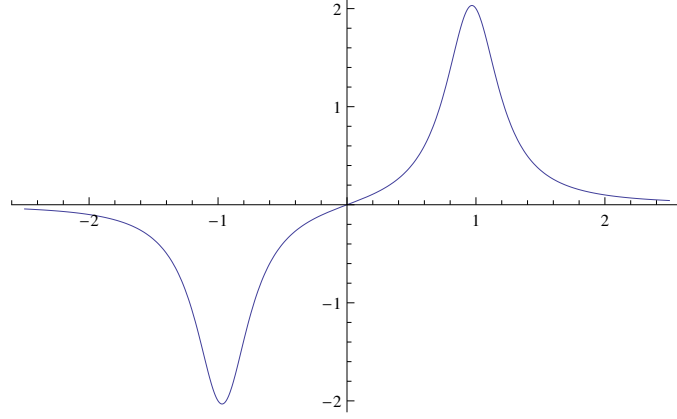


Figure 12: The imaginary, dissipative part of the response function for the underdamped harmonic oscillator, plotted with $\omega_0 = 1$ and $\gamma = 0.5$.

- **Underdamped:** $\omega_0^2 > \gamma^2/4$. In this case, the poles have both a real and imaginary part. They both sit on the lower half plane. This is in agreement with our general lesson of causality which tells us that the response function must be analytic in the upper-half plane
- **Overdamped:** $\omega_0^2 < \gamma^2/4$. Now the poles lie on the negative imaginary axis. Again, there are none in the upper-half plane, consistent with causality.

We can gain some intuition by plotting the real and imaginary part of the response function for $\omega \in \mathbf{R}$. Firstly, the real part is shown in Figure 11 where we plot

$$\text{Re}\chi(\omega) = \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \quad (4.23)$$

This is the reactive part. The higher the function, the more the system will respond to a given frequency. Notice that $\text{Re}\chi(\omega)$ is an even function, as expected.

More interesting is the dissipative part of the response function,

$$\text{Im}\chi(\omega) = \frac{\omega\gamma}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \quad (4.24)$$

This is an odd function. In the underdamped case, this is plotted in Figure 12. Notice that $\text{Im}\chi$ is proportional to γ , the coefficient of friction. The function peaks around $\pm\omega_0$, at frequencies where the system naturally vibrates. This is because this is where the system is able to absorb energy. However, as $\gamma \rightarrow 0$, the imaginary part doesn't become zero: instead it tends towards two delta functions situated at $\pm\omega_0$.

4.2.2 Dissipation

We can see directly how $\text{Im}\chi(\omega)$ is related to dissipation by computing the energy absorbed by the system. This what we used to call the work done on the system before we became all sophisticated and grown-up. It is

$$\begin{aligned}
\frac{dW}{dt} &= F(t)\dot{x}(t) \\
&= F(t)\frac{d}{dt}\int_{-\infty}^{+\infty} dt' \chi(t-t')F(t') \\
&= F(t)\int_{-\infty}^{+\infty} dt' \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} (-i\omega)e^{-i\omega(t-t')}\chi(\omega)F(t') \\
&= \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} \frac{d\omega'}{2\pi} [-i\omega\chi(\omega)]e^{-i(\omega+\omega')t}F(\omega)F(\omega')
\end{aligned} \tag{4.25}$$

Let's drive the system with a force of a specific frequency Ω , so that

$$F(t) = F_0 \cos \Omega t = F_0 \text{Re}(e^{-i\Omega t})$$

Notice that it's crucial to make sure that the force is real at this stage of the calculation because the reality of the force (or source) was the starting point for our discussion of the analytic properties of response functions in section 4.1.2. In a more pedestrian fashion, we can see that it's going to be important because our equation above is not linear in $F(\omega)$, so it's necessary to take the real part before progressing. Taking the Fourier transform, the driving force is

$$F(\omega) = 2\pi F_0 [\delta(\omega - \Omega) + \delta(\omega + \Omega)]$$

Inserting this into (4.25) gives

$$\frac{dW}{dt} = -iF_0^2\Omega [\chi(\Omega)e^{-i\Omega t} - \chi(-\Omega)e^{+i\Omega t}] [e^{-i\Omega t} + e^{i\Omega t}] \tag{4.26}$$

This is still oscillating with time. It's more useful to take an average over a cycle,

$$\overline{\frac{dW}{dt}} \equiv \frac{\Omega}{2\pi} \int_0^{2\pi/\Omega} dt \frac{dW}{dt} = -iF_0^2\Omega [\chi(\Omega) - \chi(-\Omega)]$$

But we've already seen that $\text{Re}\chi(\omega)$ is an even function, while $\text{Im}\chi(\omega)$ is an odd function. This allows us to write

$$\overline{\frac{dW}{dt}} = 2F_0^2\Omega \text{Im}\chi(\Omega) \tag{4.27}$$

We see that the work done is proportional to $\text{Im}\chi$. To derive this result, we didn't need the exact form of the response function; only the even/odd property of the real/imaginary parts, which follow on general grounds. For our damped harmonic oscillator, we can now use the explicit form (4.24) to derive

$$\frac{d\overline{W}}{dt} = 2F_0^2 \frac{\gamma\Omega^2}{(\omega_0^2 - \Omega^2)^2 + (\gamma\Omega)^2}$$

This is a maximum when we shake the harmonic oscillator at its natural frequency, $\Omega = \omega_0$. As this example illustrates, the imaginary part of the response function tells us the frequencies at which the system naturally vibrates. These are the frequencies where the system can absorb energy when shaken.

4.2.3 Hydrodynamic Response

For our final classical example, we'll briefly return to the topic of hydrodynamics. One difference with our present discussion is that the dynamical variables are now functions of both space and time. A typical example that we'll focus on here is the mass density, $\rho(\vec{x}, t)$. Similarly, the driving force (or, in the context of quantum mechanics, the source) is similarly a function of space and time.

Rather than playing at the full Navier-Stokes equation, here we'll instead just look at a simple model of diffusion. The continuity equation is

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$$

We'll write down a simple model for the current,

$$\vec{J} = -D\vec{\nabla}\rho + \vec{F} \tag{4.28}$$

where D is the diffusion constant and the first term gives rise to Fick's law that we met already in Section 1. The second term, $\vec{F} = \vec{F}(\vec{x}, t)$, is the driving force. Combining this with the continuity equation gives,

$$\frac{\partial \rho}{\partial t} - D\nabla^2 \rho = -\vec{\nabla} \cdot \vec{F} \tag{4.29}$$

We want to understand the response functions associated to this force. This includes both the response of ρ and the response of $\vec{J}$,

For simplicity, let's work in a single spatial dimension so that we can drop the vector indices. We write

$$\begin{aligned}\rho(x, t) &= \int dx' dt' \chi_{\rho J}(x', t'; x, t) F(x', t) \\ J(x, t) &= \int dx' dt' \chi_{JJ}(x', t'; x, t) F(x', t)\end{aligned}$$

where we've called the second label J on both of these functions to reflect the fact that F is a driving force for J . We follow our discussion of Section 4.1.1. We now assume that our system is invariant under both time and space translations which ensures that the response function depend only on $t' - t$ and $x' - x$. We then Fourier transform with respect to both time and space. For example,

$$\rho(\omega, k) = \int dx dt e^{i(\omega t - kx)} \rho(x, t)$$

Then in momentum and frequency space, the response functions become

$$\begin{aligned}\rho(\omega, k) &= \chi_{\rho J}(\omega, k) F(\omega, k) \\ J(\omega, k) &= \chi_{JJ}(\omega, k) F(\omega, k)\end{aligned}$$

The diffusion equation (4.29) immediately gives an expression for $\chi_{\rho J}$. Substituting the resulting expression into (4.28) then gives us χ_{JJ} . The response functions are

$$\chi_{\rho J} = \frac{ik}{-i\omega + Dk^2} \quad , \quad \chi_{JJ} = \frac{-i\omega}{-i\omega + Dk^2}$$

Both of the denominator have poles on the imaginary axis at $\omega = -iDk^2$. This is the characteristic behaviour of response functions capturing diffusion.

Our study of hydrodynamics in Sections 2.4 and 2.5 revealed a different method of transport, namely sound. For the ideal fluid of Section 2.4, the sound waves travelled without dissipation. The associated response function has the form

$$\chi_{\text{sound}} \sim \frac{1}{\omega^2 - v_s^2 k^2}$$

which is simply the Green's function for the wave equation. If one includes the effect of dissipation, the poles of the response function pick up a (negative) imaginary part. For sound waves in the Navier-Stokes equation, we computed the location of these poles in (2.76).

4.3 Quantum Mechanics and the Kubo Formula

Let's now return to quantum mechanics. Recall the basic set up: working in the Heisenberg picture, we add to a Hamiltonian the perturbation

$$H_{\text{source}}(t) = \phi_j(t) \mathcal{O}_j(t) \quad (4.30)$$

where there is an implicit sum over j , labelling the operators in the theory and, correspondingly, the different sources that we can turn on. Usually in any given situation we only turn on a source for a single operator, but we may be interested in how this source affects the expectation value of any other operator in the theory, $\langle \mathcal{O}_i \rangle$. However, if we restrict to small values of the source, we can address this using standard perturbation theory. We introduce the time evolution operator,

$$U(t, t_0) = T \exp \left(-i \int_{t_0}^t H_{\text{source}}(t') dt' \right)$$

which is constructed to obey the operator equation $idU/dt = H_{\text{source}}U$. Then, switching to the interaction picture, states evolve as

$$|\psi(t)\rangle_I = U(t, t_0) |\psi(t_0)\rangle_I$$

We'll usually be working in an ensemble of states described by a density matrix ρ . If, in the distant past $t \rightarrow -\infty$, the density matrix is given by ρ_0 , then at some finite time it evolves as

$$\rho(t) = U(t) \rho_0 U^{-1}(t)$$

with $U(t) = U(t, t_0 \rightarrow -\infty)$. From this we can compute the expectation value of any operator $\mathcal{O}_j$ in the presence of the sources ϕ . Working to first order in perturbation theory (from the third line below), we have

$$\begin{aligned} \langle \mathcal{O}_i(t) \rangle|_{\phi} &= \text{Tr } \rho(t) \mathcal{O}_i(t) \\ &= \text{Tr } \rho_0(t) U^{-1}(t) \mathcal{O}_i(t) U(t) \\ &\approx \text{Tr } \rho_0(t) \left(\mathcal{O}_i(t) + i \int_{-\infty}^t dt' [H_{\text{source}}(t'), \mathcal{O}_i(t)] + \dots \right) \\ &= \langle \mathcal{O}_i(t) \rangle|_{\phi=0} + i \int_{-\infty}^t dt' \langle [H_{\text{source}}(t'), \mathcal{O}_i(t)] \rangle + \dots \end{aligned}$$

Inserting our explicit expression for the source Hamiltonian gives the change in the expectation value, $\delta \langle \mathcal{O}_i \rangle = \langle \mathcal{O}_i \rangle_{\phi} - \langle \mathcal{O}_i \rangle_{\phi=0}$,

$$\begin{aligned} \delta \langle \mathcal{O}_i \rangle &= i \int_{-\infty}^t dt' \langle [\mathcal{O}_j(t'), \mathcal{O}_i(t)] \rangle \phi_j(t') \\ &= i \int_{-\infty}^{+\infty} dt' \theta(t-t') \langle [\mathcal{O}_j(t'), \mathcal{O}_i(t)] \rangle \phi_j(t') \end{aligned} \quad (4.31)$$

where, in the second line, we have done nothing more than use the step function to extend the range of the time integration to $+\infty$. Comparing this to our initial definition given in (4.4), we see that the response function in a quantum theory is given by the two-point function,

$$\chi_{ij}(t - t') = -i\theta(t - t') \langle [\mathcal{O}_i(t), \mathcal{O}_j(t')] \rangle \quad (4.32)$$

This important result is known as the *Kubo formula*. (Although sometimes the name “Kubo formula” is restricted to specific examples of this equation which govern transport properties in quantum field theory. We will derive these examples in Section 4.4).

4.3.1 Dissipation Again

Before we make use of the Kubo formula, we will first return to the question of dissipation. Here we repeat the calculation of 4.2.2 where we showed that, for classical systems, the energy absorbed by a system is proportional to $\text{Im } \chi$. Here we do the same for quantum systems. The calculation is a little tedious, but worth ploughing through.

As in the classical context, the work done is associated to the change in the energy of the system which, this time, can be written as

$$\frac{dW}{dt} = \frac{d}{dt} \text{Tr } \rho H = \text{Tr}(\dot{\rho} H + \rho \dot{H})$$

To compute physical observables, it doesn’t matter if we work in the Heisenberg or Schrödinger picture. So let’s revert momentarily back to the Schrödinger picture. Here, the density matrix evolves as $i\dot{\rho} = [H, \rho]$, so the first term above vanishes. Meanwhile, the Hamiltonian H changes because we’re sitting there playing around with the source (4.30), providing an explicit time dependence. To simplify our life, we’ll assume that we turn on just a single source, ϕ . Then, in the Schrödinger picture

$$\dot{H} = \mathcal{O}\dot{\phi}(t)$$

This gives us the energy lost by the system,

$$\frac{dW}{dt} = \text{Tr}(\rho \mathcal{O}\dot{\phi}) = \langle \mathcal{O} \rangle_{\phi} \dot{\phi} = [\langle \mathcal{O} \rangle_{\phi=0} + \delta \langle \mathcal{O} \rangle] \dot{\phi}$$

We again look at a periodically varying source which we write as

$$\phi(t) = \text{Re}(\phi_0 e^{-i\Omega t})$$

and we again compute the average work done over a complete cycle

$$\overline{\frac{dW}{dt}} = \frac{\Omega}{2\pi} \int_0^{2\pi/\Omega} dt \frac{dW}{dt}$$

The term $\langle \mathcal{O}(\vec{x}) \rangle_0$ cancels out when integrated over the full cycle. This leaves us with

$$\begin{aligned}
\frac{\overline{dW}}{dt} &= \frac{\Omega}{2\pi} \int_0^{2\pi/\Omega} dt \int_{-\infty}^{+\infty} dt' \chi(t-t') \phi(t') \dot{\phi}(t) \\
&= \frac{\Omega}{2\pi} \int_0^{2\pi/\Omega} dt \int_{-\infty}^{+\infty} dt' \int \frac{d\omega}{2\pi} \chi(\omega) e^{-i\omega(t-t')} \\
&\quad \times \frac{1}{4} \left[\phi_0 e^{-i\Omega t'} + \phi_0^* e^{+i\Omega t'} \right] \left[-i\Omega \phi_0 e^{-i\Omega t} + i\Omega \phi_0^* e^{+i\Omega t} \right] \\
&= \frac{1}{4} [\chi(\Omega) - \chi(-\Omega)] |\phi_0|^2 i\Omega
\end{aligned}$$

where the ϕ^2 and ϕ^{*2} terms have canceled out after performing the $\int dt$. Continuing, we only need the fact that the real part of χ is even while the imaginary part is odd. This gives us the result

$$\frac{\overline{dW}}{dt} = \frac{1}{2} \Omega \chi''(\Omega) |\phi_0|^2 \quad (4.33)$$

Finally, this calculation tells us about another property of the response function. If we perform work on a system, the energy should increase. This translates into a positivity requirement $\Omega \chi''(\Omega) \geq 0$. More generally, the requirement is that $\Omega \chi''_{ij}(\Omega)$ is a positive definite matrix.

Spectral Representation

In the case of the damped harmonic oscillator, we saw explicitly that the dissipation was proportional to the coefficient of friction, γ . But for our quantum systems, the dynamics is entirely Hamiltonian: there is no friction. So what is giving rise to the dissipation? In fact, the answer to this can also be found in our analysis of the harmonic oscillator, for there we found that in the limit $\gamma \rightarrow 0$, the dissipative part of the response function χ'' doesn't vanish but instead reduces to a pair of delta functions. Here we will show that a similar property holds for a general quantum system.

We'll take the state of our quantum system to be described by a density matrix describing the canonical ensemble, $\rho = e^{-\beta H}$. Taking the Fourier transform of the Kubo formula (4.32) gives

$$\chi_{ij}(\omega) = -i \int_0^\infty dt e^{i\omega t} \text{Tr} \left(e^{-\beta H} [\mathcal{O}_i(t), \mathcal{O}_j(0)] \right)$$

We will need to use the fact that operators evolve as $\mathcal{O}_i(t) = U^{-1} \mathcal{O}_i(0) U$ with $U = e^{-iHt}$ and will evaluate $\chi_{ij}(\omega)$ by inserting complete basis of energy states

$$\chi_{ij}(\omega) = -i \int_0^\infty dt e^{i\omega t} \sum_{mn} e^{-E_m \beta} [\langle m | \mathcal{O}_i | n \rangle \langle n | \mathcal{O}_j | m \rangle e^{i(E_m - E_n)t}]$$

$$-\langle m|\mathcal{O}_j|n\rangle\langle n|\mathcal{O}_i|m\rangle e^{i(E_n-E_m)t}]$$

To ensure that the integral is convergent for $t > 0$, we replace $\omega \rightarrow \omega + i\epsilon$. Then performing the integral over $\int dt$ gives

$$\begin{aligned}\chi_{ij}(\omega + i\epsilon) &= \sum_{m,n} e^{-E_m\beta} \left[\frac{(\mathcal{O}_i)_{mn}(\mathcal{O}_j)_{nm}}{\omega + E_m - E_n + i\epsilon} - \frac{(\mathcal{O}_j)_{mn}(\mathcal{O}_i)_{nm}}{\omega + E_n - E_m + i\epsilon} \right] \\ &= \sum_{m,n} \frac{(\mathcal{O}_i)_{mn}(\mathcal{O}_j)_{nm}}{\omega + E_m - E_n + i\epsilon} (e^{-E_m\beta} - e^{-E_n\beta})\end{aligned}$$

which tells us that the response function has poles just below the real axis,

$$\omega = E_n - E_m - i\epsilon$$

Of course, we knew on general grounds that the poles couldn't lie in the upper half-plane: we see that in a Hamiltonian system the poles lie essentially on the real axis (as $\epsilon \rightarrow 0$) at the values of the frequency that can excite the system from one energy level to another. In any finite quantum system, we have an isolated number of singularities.

As in the case of the harmonic oscillator, in the limit $\epsilon \rightarrow 0$, the imaginary part of the response function doesn't disappear: instead it becomes a sum of delta function spikes

$$\chi'' \sim \sum_{m,n} \frac{\epsilon}{(\omega + E_m - E_n)^2 + \epsilon^2} \longrightarrow \sum_{m,n} \delta(\omega - (E_n - E_m))$$

The expression above is appropriate for quantum systems with discrete energy levels. However, in infinite systems — and, in particular, in the quantum field theories that we turn to shortly — these spikes can merge into smooth functions and dissipative behaviour can occur for all values of the frequency.

4.3.2 Fluctuation-Dissipation Theorem

We have seen above that the imaginary part of the response function governs the dissipation in a system. Yet, the Kubo formula (4.32) tells us that the response formula can be written in terms of a two-point correlation function in the quantum theory. And we know that such two-point functions provide a measure of the variance, or fluctuations, in the system. This is the essence of the fluctuation-dissipation theorem which we'll now make more precise.

First, the form of the correlation function in (4.32) — with the commutator and funny theta term — isn't the simplest kind of correlation we could image. The more basic correlation function is simply

$$S_{ij}(t) \equiv \langle \mathcal{O}_i(t) \mathcal{O}_j(0) \rangle$$

where we have used time translational invariance to set the time at which $\mathcal{O}_j$ is evaluated to zero. The Fourier transform of this correlation function is

$$S_{ij}(\omega) = \int dt e^{i\omega t} S_{ij}(t) \quad (4.34)$$

The content of the fluctuation-dissipation theorem is to relate the dissipative part of the response function to the fluctuations $S(\omega)$ in the vacuum state which, at finite temperature, means the canonical ensemble $\rho = e^{-\beta H}$.

There is a fairly pedestrian proof of the theorem using spectral decomposition (i.e. inserting a complete basis of energy eigenstates as we did in the previous section). Here we instead give a somewhat slicker proof although, as we will see, it requires us to do something fishy somewhere. We proceed by writing an expression for the dissipative part of the response function using the Kubo formula (4.32),

$$\begin{aligned} \chi''_{ij}(t) &= -\frac{i}{2} [\chi_{ij}(t) - \chi_{ji}(-t)] \\ &= -\frac{1}{2} \theta(t) \left[\langle \mathcal{O}_i(t) \mathcal{O}_j(0) \rangle - \langle \mathcal{O}_j(0) \mathcal{O}_i(t) \rangle \right] \\ &\quad + \frac{1}{2} \theta(-t) \left[\langle \mathcal{O}_j(-t) \mathcal{O}_i(0) \rangle - \langle \mathcal{O}_i(0) \mathcal{O}_j(-t) \rangle \right] \end{aligned}$$

By time translational invariance, we know that $\langle \mathcal{O}_j(0) \mathcal{O}_i(t) \rangle = \langle \mathcal{O}_j(-t) \mathcal{O}_i(0) \rangle$. This means that the step functions arrange themselves to give $\theta(t) + \theta(-t) = 1$, leaving

$$\chi''_{ij}(t) = -\frac{1}{2} \langle \mathcal{O}_i(t) \mathcal{O}_j(0) \rangle + \frac{1}{2} \langle \mathcal{O}_j(-t) \mathcal{O}_i(0) \rangle \quad (4.35)$$

But we can re-order the operators in the last term. To do this, we need to be sitting in the canonical ensemble, so that the expectation value is computed with respect to the Boltzmann density matrix. We then have

$$\begin{aligned} \langle \mathcal{O}_j(-t) \mathcal{O}_i(0) \rangle &= \text{Tr} e^{-\beta H} \mathcal{O}_j(-t) \mathcal{O}_i(0) \\ &= \text{Tr} e^{-\beta H} \mathcal{O}_j(-t) e^{\beta H} e^{-\beta H} \mathcal{O}_i(0) \\ &= \text{Tr} e^{-\beta H} \mathcal{O}_i(0) \mathcal{O}_j(-t + i\beta) \\ &= \langle \mathcal{O}_i(t - i\beta) \mathcal{O}_j(0) \rangle \end{aligned}$$

The third line above is where we've done something slippery: we've treated the density matrix $\rho = e^{-\beta H}$ as a time evolution operator, but one which evolves the operator in the imaginary time direction! In the final line we've used time translational invariance, now both in real and imaginary time directions. While this may look dodgy, we can turn it into something more palatable by taking the Fourier transform. The dissipative part of the response function can be written in terms of correlation functions as

$$\chi''_{ij}(t) = -\frac{1}{2} \left[\langle \mathcal{O}_i(t) \mathcal{O}_j(0) \rangle - \langle \mathcal{O}_i(t - i\beta) \mathcal{O}_j(0) \rangle \right] \quad (4.36)$$

Taking the Fourier transform then gives us our final expression:

$$\chi''_{ij}(\omega) = -\frac{1}{2} [1 - e^{-\beta\omega}] S_{ij}(\omega) \quad (4.37)$$

This is the fluctuation-dissipation theorem, relating the fluctuations in frequency space, captured by $S(\omega)$, to the dissipation, captured by $\chi''(\omega)$. Indeed, a similar relationship holds already in classical physics; the most famous example is the Einstein relation that we met in Section 3.1.3.

The physics behind (4.37) is highlighted a little better if we invert the equation. We can write

$$S_{ij}(\omega) = -2 [n_B(\omega) + 1] \chi''_{ij}(\omega)$$

where $n_B(\omega) = (e^{\beta\omega} - 1)^{-1}$ is the Bose-Einstein distribution function. Here we see explicitly the two contributions to the fluctuations: the $n_B(\omega)$ factor is due to thermal effects; the “+1” can be thought of as due to inherently quantum fluctuations. As usual, the classical limit occurs for high temperatures with $\beta\omega \ll 1$ where $n_B(\omega) \approx k_B T / \omega$. In this regime, the fluctuation dissipation theorem reduces to its classical counterpart

$$S_{ij}(\omega) = -\frac{2k_B T}{\omega} \chi''_{ij}(\omega)$$

4.4 Response in Quantum Field Theory

We end these lectures by describing how response theory can be used to compute some of the transport properties that we've encountered in previous sections. To do this, we work with [Quantum Field Theory](#) where the operators become functions of space and time, $\mathcal{O}(\vec{x}, t)$. In the context of condensed matter, this is the right framework to describe many-body physics. In the context of particle physics, this is the right framework to describe everything.

Suppose that you take a quantum field theory, place it in a state with a finite amount of stuff (whatever that stuff is) and heat it up. What is the right description of the resulting dynamics? From our earlier discussion, we know the answer: the low-energy excitations of the system are described by hydrodynamics, simply because this is the universal description that applies to everything. (Actually, we're brushing over something here: the exact form of the hydrodynamics depends on the symmetries of the theory, both broken and unbroken). All that remains is to identify the transport coefficients, such as viscosity and thermal conductivity, that arise in the hydrodynamic equations. But how to do that starting from the quantum field?

The answer to this question lies in the machinery of linear response that we developed above. For a quantum field, we again add source terms to the action, now of the form

$$H_{\text{source}}(t) = \int d^{d-1}\vec{x} \phi_i(\vec{x}, t) \mathcal{O}_i(\vec{x}, t) \quad (4.38)$$

The response function χ is again defined to be the change of the expectation values of $\mathcal{O}$ in the presence of the source ϕ ,

$$\delta\langle\mathcal{O}_i(\vec{x}, t)\rangle = \int d^d\vec{x}' dt' \chi_{ij}(\vec{x}, t; \vec{x}', t') \phi_j(\vec{x}', t') \quad (4.39)$$

All the properties of the response function that we derived previously also hold in the context of quantum field theory. Indeed, for the most part, the label $\vec{x}$ and $\vec{x}'$ can be treated in the same way as the label i, j . Going through the steps leading to the Kubo formula (4.32), we now find

$$\chi_{ij}(\vec{x}, \vec{x}'; t - t') = -i\theta(t - t') \langle [\mathcal{O}_i(\vec{x}, t), \mathcal{O}_j(\vec{x}', t')] \rangle \quad (4.40)$$

We learned in our first course on [Quantum Field Theory](#) that the two-point functions are Green's functions. Usually, when thinking about scattering amplitudes, we work with time-ordered (Feynman) correlation functions that are relevant for building perturbation theory. Here, we are interested in the *retarded* correlation functions, characterised by the presence of the step function sitting in front of (4.40).

Finally, if the system exhibits translational invariance in both space and time, then the response function depends only on the differences $t - t'$ and $\vec{x} - \vec{x}'$. In this situation it is useful to work in momentum and frequency space, so that the (4.39) becomes

$$\delta\langle\mathcal{O}_i(\vec{k}, \omega)\rangle = \chi_{ij}(\vec{k}, \omega) \phi_j(\vec{k}, \omega) \quad (4.41)$$

Electrical Conductivity

Consider a quantum field theory with a $U(1)$ global symmetry. By Noether's theorem, there is an associated conserved current $J^\mu = (J^0, J^i)$, obeying $\partial_\mu J^\mu = 0$. This current is an example of a composite operator. It couples to a source which is a gauge field $A_\mu(x)$,

$$H_{\text{source}} = \int d^{d-1}\vec{x} A_\mu J^\mu \quad (4.42)$$

Here A_μ is the background gauge field of electromagnetism. However, for the purposes of our discussion, we do not take A_μ to have dynamics of its own. Instead, we treat it as a fixed source, under our control.

There is, however, a slight subtlety. In the presence of the background gauge field, the current itself may be altered so that it depends on A_μ . A simple, well known, example of this occurs for a free, relativistic, complex scalar field φ . The conserved current in the presence of the background field is given by

$$J^\mu = ie[\varphi^\dagger \partial^\mu \varphi - (\partial^\mu \varphi^\dagger) \varphi] - e^2 A^\mu \varphi^\dagger \varphi \quad (4.43)$$

where e is the electric charge. With this definition, the Lagrangian can be written in terms of covariant derivatives $\mathcal{D}_\mu \varphi = \partial_\mu \varphi - ie A_\mu \varphi$,

$$L = \int d^{d-1}\vec{x} |\partial_\mu \varphi|^2 + A_\mu J^\mu = \int d^{d-1}\vec{x} |\mathcal{D}_\mu \varphi|^2 \quad (4.44)$$

For non-relativistic fields (either bosons or fermions), similar A_μ terms arise in the current for the spatial components.

We want to derive the response of the system to a background electric field. Which, in more basic language, means that we want to derive Ohm's law in our quantum field theory. This is

$$\langle J_i(\vec{k}, \omega) \rangle = \sigma_{ij}(\vec{k}, \omega) E_j(\vec{k}, \omega) \quad (4.45)$$

Here E_i is the background electric field in Fourier space and σ_{ij} is the conductivity tensor. In a system with rotational and parity invariance (which, typically means in the absence of a magnetic field) we have $\sigma_{ij} = \sigma \delta_{ij}$, so that the current is parallel to the applied electric field. Here we will work with the more general case. Our goal is to get an expression for σ_{ij} in terms of correlation functions in the field theory. Applying (4.41) with the perturbation (4.42), we have

$$\delta \langle J_\mu \rangle = \langle J_\mu \rangle - \langle J_\mu \rangle_0 = -i \int_{-\infty}^t dt' d^3\vec{x} \langle [J_\mu(\vec{x}, t), J_\nu(\vec{x}', t')] \rangle_0 A_\nu(\vec{x}', t') \quad (4.46)$$

The subscript 0 here means the quantum average in the state $A_\mu = 0$ before we turn on the background field. Let's start by looking at the term $\langle J_i \rangle_0$. You might think that there are no currents before we turn on the background field. But, in fact, the extra term in (4.43) gives a contribution even if – as we'll assume – the unperturbed state has no currents. This contribution is

$$\langle J_i \rangle_0 = e^2 A_i \langle \varphi^\dagger \varphi \rangle_0 = e A_i \rho$$

where ρ is the background charge density. Notice it is not correct to set $A_i = 0$ in this expression; the subscript 0 only means that we are evaluating the expectation value in the $A_i = 0$ quantum state.

Let's now deal with the right-hand side of (4.46). If we work in $A_0 = 0$ gauge (where things are simple), the electric field is given by $E_i = -\dot{A}_i$. In Fourier transform space, this becomes

$$A_i(\omega) = \frac{E_i(\omega)}{i\omega} \quad (4.47)$$

We can now simply Fourier transform (4.46) to get it in the form of Ohm's law (4.45). The conductivity tensor has two contributions: the first from the background charge density; the second from the retarded Green's function

$$\sigma_{ij} = -\frac{e\rho}{i\omega} \delta_{ij} + \frac{\chi_{ij}(\vec{k}, \omega)}{i\omega} \quad (4.48)$$

with the Fourier transform of the retarded Green's function given in terms of the current-current correlation function

$$\chi_{ij}(\vec{k}, \omega) = -i \int_{-\infty}^{\infty} dt d^3\vec{x} \theta(t) e^{i(\omega t - \vec{k} \cdot \vec{x})} \langle [J_i(\vec{x}, t), J_j(\vec{0}, 0)] \rangle$$

This is the Kubo formula for conductivity.

Viscosity

We already saw in Section 2 that viscosity is associated to the transport of momentum. And, just as for electric charge, momentum is conserved. For field theories that are invariant under space and time translations, Noether's theorem gives rise to four currents, associated to energy and momentum conservation. These are usually packaged together into the stress-energy tensor $T^{\mu\nu}$, obeying $\partial_\mu T^{\mu\nu} = 0$. (We already met this object in a slightly different guise in Section 2, where the spatial components appeared as the pressure tensor P_{ij} and the temporal components as the overall velocity u_i).

The computation of viscosity in the framework of quantum field theory is entirely analogous to the computation of electrical conductivity. The electric current is simply replaced by the momentum current. Indeed, as we already saw in Section 2.5.3, the viscosity tells us the ease with which momentum in, say, the x -direction can be transported in the z -direction. For such a set-up, the relevant component of the current is T^{xz} . The analog of the formula for electrical conductivity can be re-interpreted as a formula for viscosity. There are two differences. Firstly, there is no background charge density. Secondly, the viscosity is for a constant force, meaning that we should take the $\omega \rightarrow 0$ and $\vec{k} \rightarrow 0$ limit of our equation. We have

$$\chi_{xz,xz}(\vec{k}, \omega) = -i \int_{-\infty}^{\infty} dt d^3\vec{x} \theta(t) e^{i(\omega t - \vec{k} \cdot \vec{x})} \langle [T_{xz}(\vec{x}, t), T_{xz}(\vec{0}, 0)] \rangle$$

and

$$\eta = \lim_{\omega \rightarrow 0} \frac{\chi_{xz,xz}(0, \omega)}{i\omega}$$

This is the Kubo formula for viscosity.